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Hamiltonian formulation of guiding center motion

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A Hamiltonian theory of guiding center motion which uses rectangular coordinates in physical space and noncanonical coordinates in phase space is presented. The averaging methods preserve two important features of Hamiltonian systems, viz., conservation of energy (for time-independent fields) and Liouville's theorem. These features are sacrificed by the traditional averaging methods. The methods also relieve much of the burden of higher order perturbation calculations, and the drift equations for fully electromagnetic fields are extended to one higher order than they have been known in the past. The first correction to the relativistic magnetic moment is also calculated. Many applications are anticipated, both to single particle motion and to kinetic theory.

I. INTRODUCTION

In a recent paper I have described the mathematical apparatus of a new approach to a Hamiltonian formulation of guiding center motion, and I have illustrated the method with motion in the field $\mathbf{B} = B(x, y)\hat{z}$. In this paper I will extend those results to nonrelativistic guiding center motion in arbitrary electromagnetic fields.

Without going into detail, we may roughly outline the strategy of Ref. 1 and this paper. An examination of the requirements of Hamiltonian perturbation theory reveals that one does not have to employ strictly canonical coordinates in phase space. In the first step of this work we deal more with Poisson bracket relations than with averaging methods in the process of finding a set of phase space coordinates to which Hamiltonian perturbation methods can be applied. The resulting coordinates are the traditional guiding center position $\mathbf{X}$, the parallel velocity U, the kinetic energy K, and the magnetic moment M, with higher order, oscillatory terms. It is interesting that the requirements of our Poisson bracket relations alone lead to these familiar variables. The coordinate transformation which produces these guiding center variables is called the Darboux transformation, and it is discussed in Sec. III.

The second step of this work is to carry out an averaging transformation, using traditional Hamiltonian methods (in this case, Lie transforms). The averaging transformation removes the higher order oscillatory behavior of the guiding center variables and produces drift equations. This transformation is discussed in Sec. IV.

The drift equations are carried to one order higher than they have traditionally been known, i.e., through $O(\epsilon)$ in the parallel motion and through $O(\epsilon^2)$ in the perpendicular motion. For the special case of purely magnetostatic fields, the drift equations have been carried to the same order by Northrop and Rome,[2] using non-Hamiltonian methods. When restricted to purely magnetostatic fields, our results agree with theirs, although our equations of motion are somewhat simpler. Because there is interest in the drift equations, apart from the methods used to derive them, we have summarized these equations in Appendix D.

We feel, however, that the methods of this paper have broader applicability to plasma physics than simply the derivation of drift equations, and for this reason we have described them in detail. Hamiltonian methods are well-suited to certain applications in kinetic theory,[3-8] and the range of applications continues to widen.[9] Our methods should be especially valuable for treating higher order effects in the ratio of gyroradius to scale length.

We have also applied our methods to relativistic guiding center motion. Since this case is of lesser interest to plasma physics, we have simply summarized, in Appendix E, the one result of this calculation which is of greatest interest, namely, the $O(\epsilon)$ term of the magnetic moment series. This term seems not to have been derived before.

Our approach is but one of several known Hamiltonian methods for treating the guiding center expansion. The earliest was devised by Gardner,[10] who used field line coordinates and mixed-variable generating functions. These methods have been used in applications,[11] and recently Weitzner[12] has used them to study guiding center motion in nearly axisymmetric fields.

Recently, Mynick[13] has developed a theory of guiding center motion using Hamiltonian methods. He also uses field line coordinates and mixed-variable generating functions, but his approach differs from Gardner's in that he uses Lie transforms as well. He also treats the perpendicular and parallel scale lengths with different ordering parameters. His results are perturbative, i.e., represented by power series, only in the parallel ordering parameter.

This work uses rectangular coordinates in configuration space, rather than field line coordinates. The use of rectangular coordinates seems to compel one to employ noncanonical coordinates in phase space. Although noncanonical coordinates are unfamiliar, they are easily understood if one uses only the abstract algebra of Poisson brackets, and refrains from explicit representations in terms of q's and p's. A relatively simple application of this principle is given in Sec. IIIF,

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where we derive the familiar drift equations with the Poisson bracket formalism. Greater detail concerning the mathematical theory of noncanonical variables may be found in Refs. 1, 14, and 15.

In order to use Hamiltonian methods in a noncanonical coordinate system, one must know the Poisson brackets of the coordinates among themselves. For the guiding center variables used in this work, these Poisson brackets are given in Eqs. (33). They have a number of interesting properties, among which are the fact that they are rational functions (not power series) in the adiabatic parameter ϵ, and that they are valid to all orders, i.e., they are not truncated power series.

Because the Poisson brackets of the guiding center variables among themselves can be expressed in closed form, it is possible to write down guiding center equations which conserve both phase volume and (for time-independent fields) energy. This issue is discussed in Sec. IIIG, where we display such drift equations and compare them with the usual drift equations [see Eqs. (55)–(56)].

The organization of this paper is as follows. In Sec. II we introduce a set of variables called "physical particle variables," which describe the instantaneous physical state of the particle. In Sec. III we carry out the Darboux transformation, which produces a set of guiding center variables which have appropriate Poisson bracket relations among themselves. These variables satisfy a number of exact relations, concerning their Poisson brackets, Liouville's theorem, and the like. In Sec. IV we carry out the averaging transformation, which produces an averaged set of variables, an averaged Hamiltonian, and drift equations. We give a geometrical interpretation of some of the terms of the Hamiltonian, and we discover a divergence free vector **N** of geometrical origin which at higher order behaves like the physical magnetic field. In Sec. V we discover a new set of guiding center variables with a number of interesting properties; the equations of motion of these variables are given in Appendix D. Finally, in Sec. VI, we discuss improvements and applications of the method.

II. PHYSICAL PARTICLE VARIABLES

In this section we discuss three coordinate systems in phase space. The variables comprising these coordinate systems are defined in terms of the instantaneous physical state of the particle itself (rather than the guiding center), and they will be called "physical particle variables." For each of these coordinate systems we will write down the Hamiltonian and the Poisson brackets of the coordinates among themselves.

The dimensionless parameter specifying the guiding center approximation will be denoted by ϵ. This parameter is introduced initially by replacing the electric charge e by e/ϵ, and physical results are recovered in the end by setting $\epsilon = 1$. For mathematical reasons it is useful to regard ϵ as a variable, so that one can speak of the order of an expression in terms of its behavior as $\epsilon \to 0$. We adopt the convention that the position **x** and velocity **v** of the particle are $O(\epsilon^0) = O(1)$.

The magnetic field **B** will be taken to be $O(1)$, as will the perpendicular component $\mathbf{E}_\perp$ of the electric field. It is necessary to assume, however, that $E_\parallel = O(\epsilon)$, for if $E_\parallel$ were $O(1)$, then the expansion in powers of ϵ would unavoidably contain secularities. In physical terms, the guiding center approximation breaks down when $E_\parallel = O(1)$, because velocities which are $O(1/\epsilon)$ may be reached within times which are $O(1)$, and the variation in the magnetic field as seen by the particle during a gyroperiod is then not small.

To simplify the notation, we choose units so that $m = c = |e| = 1$. This entire calculation is carried out for a positive particle, although the results may be transcribed so as to be valid for particles of any charge according to the following rule. Every occurrence of the symbols for the fields $\mathrm{A}, \phi, \mathrm{B}, B, B_{ij}, \mathbf{E}$, and the modified fields A^*, ϕ^*, etc. (but not the fields $\delta\mathrm{A}, \delta\phi$, etc.), is to be replaced by $s\mathrm{A}, s\phi$, etc., where $s = \mathrm{sgn}(e)$. Afterwards, if desired, physical units may be restored by a trivial scaling.

A. The coordinates (q,p,t,h) and (x,v,t,k_0)

In order to avoid certain difficulties with time-dependent, noncanonical coordinate transformations, we will employ an "extended phase space" of four degrees-of-freedom, with coordinates $(\mathrm{q}, \mathrm{p}, t, h)$ (see Appendix A for the special case of magnetostatic fields). In the extended phase space, the time t is treated as a phase coordinate on an equal footing with the others, and a new, energy-like variable h is introduced. This is a well-known tactic for time-dependent Hamiltonian systems.[16] The independent parameter characterizing the Hamiltonian trajectories through phase space will be denoted by τ, so that the equations of motion form a set of eight τ-independent, i.e., autonomous, differential equations.

Upon integration the differential equations of motion in the extended phase space produce eight constants. Only six constants, however, are needed to describe the physical state of the particle, so that there are two nonphysical constants of integration inherent in the use of the extended phase space. One of these merely represents the origin of time, and is of no significance. The other nonphysical constant can be taken to be the numerical value of the Hamiltonian $H(\mathrm{q}, \mathrm{p}, t, h)$ on the extended phase space. This constant enters into a physical interpretation of the variable h, which, to within the constant in question, is the total energy of the particle (kinetic plus potential).

We begin with the Hamiltonian

$$H(\mathrm{q}, \mathrm{p}, t, h) = \tfrac{1}{2}[\mathrm{p} - (1/\epsilon)\mathrm{A}(\mathrm{q}, t)]^2 + (1/\epsilon)\phi(\mathrm{q}, t) - h, \qquad (1)$$

in which the dimensionless guiding center parameter ϵ corresponds to the reciprocal of the (suppressed) electric charge e. The potentials (A, ϕ) employed in Eq. (1) can be taken in any gauge.

The variables $(\mathrm{q}, \mathrm{p}, t, h)$ satisfy the following Poisson

30 October 2026 19:36:15

bracket relations:

$$\{q_i, p_j\} = \delta_{ij}, \tag{2a}$$

$$\{t, h\} = -1. \tag{2b}$$

All other independent Poisson brackets vanish. Equation (2b) is somewhat unconventional; more commonly, one would use $-h$ in the place of h used here. One might say that our t and h are "anticonjugate" variables. Equations (2) give the components of the contravariant Poisson tensor (which was simply called the σ tensor in Ref. 1). An alternate way of writing Eqs. (2) is to give the following formula for the Poisson bracket of two phase functions F and G:

$$\{F, G\} = \frac{\partial F}{\partial \mathbf{q}} \cdot \frac{\partial G}{\partial \mathbf{p}} - \frac{\partial F}{\partial \mathbf{p}} \cdot \frac{\partial G}{\partial \mathbf{q}} - \frac{\partial F}{\partial t}\frac{\partial G}{\partial h} + \frac{\partial F}{\partial h}\frac{\partial G}{\partial t}. \tag{3}$$

Note the signs of the last two terms.

The equations of motion in the coordinates $(\mathbf{q}, \mathbf{p}, t, h)$ may easily be obtained from the Hamiltonian (1) and the Poisson brackets (2) or (3). Nevertheless, since p and h are gauge dependent, it is easier to see the meaning of the results if we first transform to a new set of gauge independent coordinates, $(\mathbf{x}, \mathbf{v}, t, k_0)$, which are given by

$$\mathbf{x} = \mathbf{q}, \tag{4a}$$

$$\mathbf{v} = \mathbf{p} - (1/\epsilon)\mathbf{A}(\mathbf{q}, t), \tag{4b}$$

$$k_0 = h - (1/\epsilon)\phi(\mathbf{q}, t). \tag{4c}$$

Physically, x is the position of the particle and v is its velocity. On the surface $H = 0$ in phase space, k_0 is the kinetic energy $\frac{1}{2}v^2$ of the particle, as may be seen from Eqs. (1) and (4). (The subscript 0 indicates that the kinetic energy is measured in the laboratory frame. The bare symbol k is reserved for later use.) Note that, by Eqs. (4b, c), p and h are $O(1/\epsilon)$. Transforming the Hamiltonian to the new coordinates gives

$$H(\mathbf{x}, \mathbf{v}, t, k_0) = \tfrac{1}{2}v^2 - k_0. \tag{5}$$

The components of the Poisson tensor in the new coordinates may be derived by using Eq. (2) or (3). The nonvanishing components are the following:

$$\{x_i, v_j\} = \delta_{ij}, \tag{6a}$$

$$\{t, k_0\} = -1, \tag{6b}$$

$$\{v_i, v_j\} = B_{ij}/\epsilon, \tag{6c}$$

$$\{\mathbf{v}, k_0\} = -\mathbf{E}/\epsilon, \tag{6d}$$

where B_{ij} is the antisymmetric tensor which is dual to the magnetic field vector B:

$$B_{ij} = \epsilon_{ijk} B_k, \tag{7}$$

where ϵ_{ijk} is the Levi-Civita symbol. (Here and throughout this paper summation over repeated indices is to be understood.) Note that the components of the Poisson tensor, like the coordinates themselves, are gauge independent in the coordinates $(\mathbf{x}, \mathbf{v}, t, k_0)$.

The Poisson brackets may be combined with the Hamiltonian (5) to obtain the equations of motion. One finds

$$\frac{d\mathbf{x}}{d\tau} = \mathbf{v}, \quad \frac{dt}{d\tau} = 1,$$

$$\frac{d\mathbf{v}}{d\tau} = \frac{1}{\epsilon}(\mathbf{v} \times \mathbf{B} + \mathbf{E}), \quad \frac{dk_0}{d\tau} = \frac{1}{\epsilon}(\mathbf{v} \cdot \mathbf{E}),$$

in accordance with the physical interpretations given for the variables $(\mathbf{x}, \mathbf{v}, t, k_0)$.

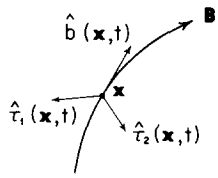


FIG. 1. The three unit vectors $\hat{b}$, $\hat{\tau}_1$, and $\hat{\tau}_2$.

B. The coordinates $(\mathbf{x}, t, k, \mu, \theta, w)$

We now introduce the new phase space coordinates $(\mathbf{x}, t, k, u, \theta, w)$, which begin to describe the gyration of the particle. The transformation $(\mathbf{x}, \mathbf{v}, t, k_0) \to (\mathbf{x}, t, k, u, \theta, w)$ is summarized in Eqs. (9); presently, we note that k is the kinetic energy of the particle, u is the parallel velocity, θ is the gyrophase, and w is the magnitude of the perpendicular velocity, all measured in a frame moving instantaneously with the $\mathbf{E} \times \mathbf{B}$ drift. These variables represent the instantaneous state of the particle, and are not averaged.

To define these variables precisely we first introduce a right-handed, orthogonal triad of unit vector fields, $(\hat{\tau}_1, \hat{\tau}_2, \hat{b})$, as shown in Fig. 1. The field $\hat{b}(\mathbf{x}, t)$ is along the magnetic field, i.e., $\hat{b} = \mathbf{B}/B$, and $\hat{\tau}_1(\mathbf{x}, t)$ and $\hat{\tau}_2(\mathbf{x}, t)$ are two more unit vector fields which span the plane perpendicular to $\hat{b}(\mathbf{x}, t)$. These unit vector fields satisfy a number of useful identities, which are summarized in Appendix B.

We let D represent the $\mathbf{E} \times \mathbf{B}$ drift velocity, and D its magnitude

$$\mathbf{D}(\mathbf{x}, t) = \mathbf{E} \times \mathbf{B}/B^2, \quad D(\mathbf{x}, t) = E_\perp/B. \tag{8}$$

The velocity of the particle as measured in a frame moving with the $\mathbf{E} \times \mathbf{B}$ drift is $\mathbf{v} - \mathbf{D}$. We define the gyrophase θ as the angle between the vector $\hat{b} \times (\mathbf{v} - \mathbf{D})$ and some reference direction $\hat{n}$ in the perpendicular plane. This angle is measured in a clockwise direction, so that $\dot{\theta}$ is positive for a positive particle. The definition of the reference direction $\hat{n}$, which specifies the origin of the gyrophase θ, requires some care.

Several perpendicular directions are suggested by the geometry of the electromagnetic field, such as $\hat{b} \cdot \nabla \hat{b}$ or $\mathbf{E}_\perp$. However, such choices for the origin of θ do not simplify further calculations and they will not be made here. Instead, we will allow the reference direction to be a free parameter of the problem for the time being. This will be done as follows. First, we imagine that the fields $\hat{\tau}_1(\mathbf{x}, t)$ and $\hat{\tau}_2(\mathbf{x}, t)$ are chosen in some manner which is arbitrary but fixed for the duration of the problem. Next, we introduce a scalar field $\psi(\mathbf{x}, t)$ in terms of which we define the reference direc-

30 October 2026 19:36:15

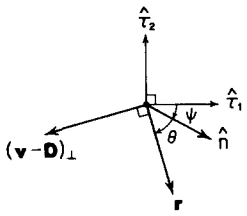


FIG. 2. The perpendicular plane. The vector $(\boldsymbol{v}-\boldsymbol{D})_\perp$ is the projection of the particle velocity, relative to the $\boldsymbol{E}\times\boldsymbol{B}$ drift velocity, and $\boldsymbol{r}=\hat{\boldsymbol{b}}\times(\boldsymbol{v}-\boldsymbol{D})_\perp$ is parallel (antiparallel) to the gyroradius vector for a positive (negative) particle. $\hat{n}$ is the local reference direction representing the origin of the gyrophase θ, and $\psi(\boldsymbol{x},t)$ specifies $\hat{n}$ relative to the axes $\hat{\tau}_1$ and $\hat{\tau}_2$.

tion $\hat{n}$

$$\hat{n}=\cos\psi\hat{\tau}_1-\sin\psi\hat{\tau}_2\,.$$

The scalar field ψ is left unspecified for now; later, we will see that there is a judicious choice for ψ which will simplify the calculations. Finally, θ is taken to be the angle between $\hat{b}\times(\mathrm{v}-\mathrm{D})$ and $\hat{n}$. Equation (9d) summarizes this discussion of the definition of the gyrophase θ, and Fig. 2 illustrates the definition geometrically.

The overall transformation $(\mathrm{x},\mathrm{v},t,k_0)\to(\mathrm{x},t,k,u,\theta,w)$ is given by

$$u=\hat{b}\cdot\mathrm{v}\,, \tag{9a}$$

$$w=[(\mathrm{v}-\mathrm{D})^2-u^2]^{1/2}\,, \tag{9b}$$

$$k=k_0-\mathrm{v}\cdot\mathrm{D}+D^2/2\,, \tag{9c}$$

$$\theta=\tan^{-1}\left(\frac{\hat{\tau}_1\cdot(\mathrm{v}-\mathrm{D})}{\hat{\tau}_2\cdot(\mathrm{v}-\mathrm{D})}\right)-\psi\,, \tag{9d}$$

where the fields $\hat{b}$, $\hat{\tau}_1$, $\hat{\tau}_2$, D, and ψ are evaluated at (x,t).

The transformation which is inverse to Eq. (9) is useful. In order to specify the inverse it is convenient to introduce two more unit vector fields, $\hat{a}$ and $\hat{c}$, which are velocity-dependent and rotate with the particle. With the coordinates $(\mathrm{x},\mathrm{v},t,k_0)$ taken as the independent variables, these unit vectors are defined by

$$\hat{c}(\mathrm{x},t,\mathrm{v})=(\mathrm{v}-\mathrm{D})_\perp/|(\mathrm{v}-\mathrm{D})_\perp|\,,\quad \hat{a}(\mathrm{x},t,\mathrm{v})=\hat{b}\times\hat{c}\,. \tag{10}$$

Thus, $\hat{c}$ is in the direction of the perpendicular velocity vector (relative to the $\mathbf{E}\times\mathbf{B}$ drift), and $\hat{a}$ is in the direction of the vector $\hat{b}\times(\mathrm{v}-\mathrm{D})$. A useful mnemonic is that $(\hat{a},\hat{b},\hat{c})$ form a right-handed triad. If we express $\hat{a}$ and $\hat{c}$ with the coordinates $(\mathrm{x},t,k,u,\theta,w)$ taken as the independent variables, we see that these vectors depend on the velocity only through the gyrophase θ,

$$\begin{aligned}\hat{a}(\mathrm{x},t,\theta)&=\cos(\theta+\psi)\hat{\tau}_1-\sin(\theta+\psi)\hat{\tau}_2\,,\\ \hat{c}(\mathrm{x},t,\theta)&=-\sin(\theta+\psi)\hat{\tau}_1-\cos(\theta+\psi)\hat{\tau}_2\,.\end{aligned} \tag{11}$$

Further properties of the vectors $\hat{a}$ and $\hat{c}$ are summarized in Appendix B.

In terms of $\hat{a}$ and $\hat{c}$, the transformation inverse to Eq. (9) is

$$\mathrm{v}=\mathrm{D}+u\hat{b}+w\hat{c}\,, \tag{12a}$$

$$k_0=k+w(\hat{c}\cdot\mathrm{D})+\tfrac{1}{2}D^2\,. \tag{12b}$$

This may be used to write the Hamiltonian in the new coordinates

$$H(\mathrm{x},t,k,u,\theta,w)=\tfrac{1}{2}(u^2+w^2)-k\,. \tag{13}$$

C. The Poisson tensor in the coordinates (x,t,k,u,θ,w)

To calculate the 28 independent components of the Poisson tensor in the coordinates $(\mathrm{x},t,k,u,\theta,w)$ is a straightforward but somewhat lengthy process. The results are summarized in Eqs. (15); here, we will outline the method used to obtain them

It is best to substitute Eqs. (12) into Eqs. (6) and to use the chain rule formula for Poisson brackets

$$\{F,G\}=\sum_{i,j}\frac{\partial F}{\partial z_i}\{z_i,z_j\}\frac{\partial G}{\partial z_j}\,, \tag{14}$$

where z_i represents the coordinates $(\mathrm{x},t,k,u,\theta,w)$, to obtain a set of linear equations for the new Poisson brackets. For example, substituting Eq. (12a) into Eq. (6a) gives

$$\delta_{ij}=\{x_i,u\}b_j+\{x_i,w\}c_j-\{x_i,\theta\}wa_j\,,$$

which is easily solved for the Poisson brackets in Eqs. 15(b)–(d). The identities of Appendix B are useful in this and subsequent steps. After dispensing with Eq. (6a) in this manner, one may proceed in the following sequence. The relation $\{t,\mathrm{v}\}=0$ implies $\{t,u\}=\{t,w\}=\{t,\theta\}=0$; Eq. (6b) implies Eq. (15e); the relation $\{\mathrm{x},k_0\}=0$ implies Eq. (15a); Eq. (6c) implies Eqs. (15g, h, k); and Eq. (6d) implies Eqs. (15f, i.j). The last two steps are the most lengthy.

The following is a summary of the results for the Poisson tensor in the coordinates $(\mathrm{x},t,k,u,\theta,w)$:

$$\{\mathrm{x},k\}=-\mathrm{D}\,, \tag{15a}$$

$$\{\mathrm{x},u\}=\hat{b}\,, \tag{15b}$$

$$\{\mathrm{x},\theta\}=-\hat{a}/w\,, \tag{15c}$$

$$\{\mathrm{x},w\}=\hat{c}\,, \tag{15d}$$

$$\{t,k\}=-1 \tag{15e}$$

$$\{k,\theta\}=(G_0+wF_0)/w\,, \tag{15f}$$

$$\{u,\theta\}=-(G_1+wF_1)/w\,, \tag{15g}$$

$$\{\theta,w\}=(B/\epsilon+G_2+wF_2)/w \tag{15h}$$

$$\{k,u\}=E_\parallel/\epsilon+G_3+wF_3\,, \tag{15i}$$

$$\{k,w\}=-(G_4+wF_4)\,, \tag{15j}$$

$$\{u,w\}=G_5+wF_5\,. \tag{15k}$$

The abbreviations F_0, F_1, etc., represent functions of (x,t,θ) which are defined in Eqs. (16), and G_0, G_1, etc., are functions of (x,t,θ,u), being simply polynomials in u, which are defined in Eqs. (17). Note that the Poisson tensor is ordered in ϵ, and that the bracket $\{\theta,w\}$ contains the dominant term, if $E_\parallel=O(\epsilon)$. The components of the Poisson tensor which are not given in Eq. (15) are zero. The equations of motion in the coordinates $(\mathrm{x},t,k,u,\theta,w)$ are easily derived from the Poisson brackets and the Hamiltonian, but they will not be needed.

Here, we summarize the definitions of the quantities

30 October 2026 19:36:15

F_0, F_1, etc. and G_0, G_1, etc. In addition to those appearing in Eq. (15), a few more are defined which will be of use later. The subscripts are merely organizational, and do not have any deep significance. In the following formula, the operator ∇ acts only on the immediately following operand. For example, $\nabla\hat{c}\cdot\hat{a}$ means $(\nabla\hat{c})\cdot\hat{a}$ and not $\nabla(\hat{c}\cdot\hat{a})$.

$$F_0=\mathbf{D}\cdot\nabla\hat{c}\cdot\hat{a}+\hat{a}\cdot\nabla\mathbf{D}\cdot\hat{c}+\hat{a}\cdot\frac{\partial\hat{c}}{\partial t}, \tag{16a}$$

$$F_1=\hat{a}\cdot\nabla\hat{b}\cdot\hat{c}+\hat{b}\cdot\nabla\hat{c}\cdot\hat{a}, \tag{16b}$$

$$F_2=\hat{c}\cdot\nabla\hat{c}\cdot\hat{a}, \tag{16c}$$

$$F_3=\mathbf{D}\cdot\nabla\hat{b}\cdot\hat{c}-\hat{b}\cdot\nabla\mathbf{D}\cdot\hat{c}+\hat{c}\cdot\frac{\partial\hat{b}}{\partial t}, \tag{16d}$$

$$F_4=\hat{c}\cdot\nabla\mathbf{D}\cdot\hat{c}, \tag{16e}$$

$$F_5=\hat{c}\cdot\nabla\hat{b}\cdot\hat{c}, \tag{16f}$$

$$F_6=\hat{a}\cdot\nabla B/B, \tag{16g}$$

$$F_7=\hat{c}\cdot\nabla B/B, \tag{16h}$$

$$F_8=\hat{a}\cdot\nabla\hat{b}\cdot\hat{a}, \tag{16i}$$

$$F_9=\hat{a}\cdot\nabla\hat{c}\cdot\hat{a}. \tag{16j}$$

$$G_0=\left(\mathbf{D}\cdot\nabla\mathbf{D}\cdot\hat{a}+\hat{a}\cdot\frac{\partial\mathbf{D}}{\partial t}\right)+u\left(\mathbf{D}\cdot\nabla\hat{b}\cdot\hat{a}-\hat{a}\cdot\nabla\hat{b}\cdot\mathbf{D}+\hat{a}\cdot\frac{\partial\hat{b}}{\partial t}\right), \tag{17a}$$

$$G_1=(\hat{a}\cdot\nabla\hat{b}\cdot\mathbf{D}+\hat{b}\cdot\nabla\mathbf{D}\cdot\hat{a})+u(\hat{b}\cdot\nabla\hat{b}\cdot\hat{a}), \tag{17b}$$

$$G_2=\hat{b}\cdot\nabla\times(\mathbf{D}+u\hat{b}), \tag{17c}$$

$$G_3=\left(\mathbf{D}\cdot\nabla\hat{b}\cdot\mathbf{D}+\mathbf{D}\cdot\frac{\partial\hat{b}}{\partial t}\right)+u(\hat{b}\cdot\nabla\hat{b}\cdot\mathbf{D}), \tag{17d}$$

$$G_4=\left(\mathbf{D}\cdot\nabla\mathbf{D}\cdot\hat{c}+\hat{c}\cdot\frac{\partial\mathbf{D}}{\partial t}\right)+u\left(\mathbf{D}\cdot\nabla\hat{b}\cdot\hat{c}-\hat{c}\cdot\nabla\hat{b}\cdot\mathbf{D}+\hat{c}\cdot\frac{\partial\hat{b}}{\partial t}\right), \tag{17e}$$

$$G_5=(\hat{b}\cdot\nabla\mathbf{D}\cdot\hat{c}+\hat{c}\cdot\nabla\hat{b}\cdot\mathbf{D})+u(\hat{b}\cdot\nabla\hat{b}\cdot\hat{c}), \tag{17f}$$

$$G_6=\left[\left(u\hat{b}\cdot\nabla+\mathbf{D}\cdot\nabla+\frac{\partial}{\partial t}\right)(u\hat{b}+\mathbf{D})\right]\cdot\hat{a}, \tag{17g}$$

$$G_7=\hat{b}\cdot(\nabla\times\mathbf{D})+2\mathbf{D}\cdot\nabla\hat{c}\cdot\hat{a}+2\hat{a}\cdot\frac{\partial\hat{c}}{\partial t}+u[\hat{b}\cdot(\nabla\times\hat{b})+2\hat{b}\cdot\nabla\hat{c}\cdot\hat{a}], \tag{17h}$$

$$G_8=(\hat{a}\cdot\nabla\mathbf{D}\cdot\hat{c}+\hat{c}\cdot\nabla\mathbf{D}\cdot\hat{a})+u(\hat{a}\cdot\nabla\hat{b}\cdot\hat{c}+\hat{c}\cdot\nabla\hat{b}\cdot\hat{a}), \tag{17i}$$

$$G_9=\left[\left(u\hat{b}\cdot\nabla+\mathbf{D}\cdot\nabla+\frac{\partial}{\partial t}\right)(u\hat{b}+\mathbf{D})\right]\cdot\hat{c}, \tag{17j}$$

$$G_{10}=(\hat{c}\cdot\nabla\mathbf{D}\cdot\hat{c}-\hat{a}\cdot\nabla\mathbf{D}\cdot\hat{a})+u(\hat{c}\cdot\nabla\hat{b}\cdot\hat{c}-\hat{a}\cdot\nabla\hat{b}\cdot\hat{a}), \tag{17k}$$

$$G_{11}=\left(\hat{b}\cdot\nabla\mathbf{D}\cdot\hat{c}+\mathbf{D}\cdot\nabla\hat{b}\cdot\hat{c}+\hat{c}\cdot\frac{\partial\hat{b}}{\partial t}\right)+2u(\hat{b}\cdot\nabla\hat{b}\cdot\hat{c}). \tag{17l}$$

D. An alternate derivation of Eqs. (15)

We conclude Sec. II with a summary of some properties of the Poisson tensor, the Lagrange tensor, and the action form which can be used to provide an alternate derivation of Eqs. (15), and which will be of use later. A deeper explanation of the mechanical significance of these objects may be found in Refs. 14 and 15.

We denote the Poisson tensor by J^{ij}. (This was called σ^{ij} in Ref. 1.) Thus, J^{ij} is the Poisson bracket $\{z_i, z_j\}$ in the given coordinate system z. Next, we define the Lagrange tensor ω_{ij} as the inverse of the Poisson tensor J^{ij}

$$J^{ik}\omega_{kj}=\delta^i_j. \tag{18}$$

(The Lagrange tensor was simply called the ω tensor in Ref. 1, and the definition there differed by a sign from that employed here.) Thus, ω_{ij} is the negative of the Lagrange bracket of z_i with z_j. Finally, the action form ρ_i is a covariant vector (i.e., a 1-form) which satisfies

$$\omega_{ij}=\frac{\partial\rho_j}{\partial z_i}-\frac{\partial\rho_i}{\partial z_j}. \tag{19}$$

That such a ρ_i exists is a consequence of the closedness of ω, i.e., $d\omega=0$. ρ is called the action form because it is closely associated with the differential of the mechanical action, $\mathbf{p}\cdot d\mathbf{q}-h dt$.

One advantage of working with ρ is that it has trivial transformation properties. This is because ρ has only one index, and that one is a covariant index. Indeed, it is possible to write down an expression for ρ which is valid in any coordinate system z, namely,

$$\rho_i=\mathbf{p}\cdot\frac{\partial\mathbf{q}}{\partial z_i}-h\frac{\partial t}{\partial z_i}. \tag{20}$$

In this expression, z represents an arbitrary coordinate system on phase space, and p, q, h, and t are considered to be expressed as functions of z.

Equation (20) is not a unique expression for the differential of the mechanical action. Rather, it is subject to the phase space analog of a gauge transformation, i.e.,

$$\rho'_i=\rho_i+\frac{\partial W}{\partial z_i}, \tag{21}$$

for any scalar W. Any two covariant vectors ρ_i, ρ'_i, connected by Eq. (21), satisfy Eq. (19) equally well, i.e., they give the same Lagrange and Poisson tensors. The scalar W is closely related to the generator of canonical transformations.

It is instructive to let $\mathbf{z}=(\mathbf{q}, \mathbf{p}, t, h)$ in Eq. (20). One immediately finds $\rho_i=(\mathbf{p}, 0, -h, 0)$, using the same ordering. This may then be substituted into Eq. (19) to find ω_{ij}, and this may in turn be inverted according to Eq. (18) to find J^{ij}. All these operations are trivial, and they yield the Poisson brackets in Eq. (2).

The same procedure can be carried out in the coordinate system $\mathbf{z}=(\mathbf{x}, t, k, u, \theta, w)$. For the action form ρ, one finds

$$\begin{aligned}\rho_{\mathbf{x}}&=(1/\epsilon)\mathbf{A}(\mathbf{x}, t)+u\hat{b}+w\hat{c}+\mathbf{D},\\ \rho_t&=-(1/\epsilon)\phi(\mathbf{x}, t)-k-w(\hat{c}\cdot\mathbf{D})-\tfrac{1}{2}D^2,\\ \rho_k&=\rho_u=\rho_\theta=\rho_w=0.\end{aligned} \tag{22}$$

The Lagrange tensor ω_{ij} is easily derived from this by using Eq. (19), but the expression is somewhat lengthy and need not be given here. Finally, inverting ω_{ij} to obtain J^{ij} provides a check on Eqs. (15). The matrix inversion is the only tedious aspect of this calculation,

30 October 2026 19:36:15

but, with the aid of Appendix C, it is still less effort than the direct procedure given previously in Sec. IIC.

III. THE DARBOUX TRANSFORMATION

The motivation for the Darboux transformation and the theory underlying it have been explained in Ref. 1, and we shall be brief where this calculation is only the natural generalization of that of Ref. 1. On the other hand, certain improvements on the methods of Ref. 1 will be employed here, and these will be explained as they arise.

A. Description of the new coordinates (X,t,K,U,θ,M)

The Darboux transformation has the form $(\mathbf{x}, t, k, u, \theta, w) \to (\mathbf{X}, t, K, U, \theta, M)$. The mathematical significance of the new variables is indicated by Eqs. (23); the following is a simple verbal description of them. The new variables are given as power series in ϵ, except for t and θ, which are exactly equal to their old counterparts. Thus, θ is still the instantaneous gyrophase. The new variables X, K, and U are equal to x, k, and u at lowest order, but they have higher order terms. These variables are not free of rapid oscillations to all orders, but they are free to one higher order than x, k, and u, due to the effects of the $O(\epsilon)$ terms. For example, X is the guiding center position through $O(\epsilon)$. The variable M (the analog of which was denoted by J in Ref. 1) is proportional to the magnetic moment at lowest order. The Darboux transformation itself is displayed in Eqs. (43), and the inverse is given in Eqs. (40).

It will be convenient for the time being to write z for the eight old variables $(\mathbf{x}, t, k, u, \theta, w)$, and Z for the six new variables $(\mathbf{X}, t, K, U)$. In terms of this notation the Darboux transformation can be written $\mathbf{z} \to (\mathbf{Z}, \theta, M)$.

One of the most interesting results of this paper is the set of Poisson brackets given in Eqs. (33), specifying the components of the Poisson tensor in the new coordinates. It should be emphasized that these are exact expressions, not power series in ϵ. These Poisson brackets are the result of one iteration of the Darboux algorithm, bringing the 8×8 Poisson tensor into $6+2$ block-diagonal form.

B. Mathematical specification of the new coordinates (X,t,K,U,θ,M)

The defining equations for the new variables Z and M, considered as functions of the old variables z, are

$$\{\mathbf{Z}, \theta\} = 0, \tag{23a}$$

$$\{\theta, M\} = 1/\epsilon, \tag{23b}$$

$$\{\mathbf{Z}, M\} = 0. \tag{23c}$$

Note that Eqs. (23a, b) are linear in the unknowns Z and M, whereas Eq. (23c) is nonlinear. We will solve the linear equations first, and then verify that the solutions satisfy the nonnlinear equations. This is a rather different approach from that taken in Ref. 1.

The characteristics of the linear differential operator appearing in Eqs. (23a, b) are the phase space curves $\mathbf{z} = \mathbf{z}(\lambda)$ which are the solutions to $d\mathbf{z}/d\lambda = \{\mathbf{z}, \theta\}$. These curves will be called the θ characteristics. It is suggestive to rewrite Eqs. (23a, b) in terms of the θ characteristics, i.e, $d\mathbf{Z}/d\lambda = 0$, $dM/d\lambda = -1/\epsilon$. Thus, M is $-1/\epsilon$ times the elapsed λ parameter, relative to some initial value surface, measured along a θ characteristic, and the quantities Z are six constants of the θ characteristics, which can be construed as coordinates on some initial value surface.

A particular initial value surface suggests itself when the equations for the θ characteristics are written out explicitly and ordered in ϵ. Calling on Eqs. (15), we have

$$\frac{d\mathbf{x}}{d\lambda}, \frac{dk}{d\lambda}, \frac{du}{d\lambda} = O(1), \tag{24a}$$

$$\frac{dt}{d\lambda} = 0 \text{ exactly}, \tag{24b}$$

$$\frac{dw}{d\lambda} = -\frac{B}{\epsilon w} + O(1). \tag{24c}$$

For small ϵ, the first term on the right-hand side of Eq. (24c) is the dominant term, and w is a monotonic function of λ, at least for neighborhoods about $w=0$ which are $O(1)$ in their linear dimensions. Within such neighborhoods, every θ characteristic meets the surface $w=0$, which we shall treat as the initial value surface. Because w is like the radial coordinate in a system of polar coordinates, the initial value surface is only six-dimensional, not seven. The gyrophase θ is not defined on the initial value surface.

We adopt the initial value conventions that $M=0$, $\mathbf{X}=\mathbf{x}$, $K=k$, and $U=u$ on the initial value surface, i.e., when $w=0$. These conventions complete the specification of the new coordinates, and hence of the Darboux transformation. [Note that by Eq. (24b), or by $\{t, \theta\}=0$, the time t is automatically a constant of the θ characteristics. Therefore, it does not change in the Darboux transformation.] A typical θ characteristic and the initial value conventions are illustrated schematically in Fig. 3.

C. The action form ρ on the initial value surface

We will defer until Sec. IIIE the explicit determination of the Darboux transformation, since the transformation is not explicitly needed for the derivation of the components of the Poisson tensor in the new coordinates. This is a fortunate circumstance, because the Darboux transformation is given by power series; indeed, we shall obtain exact, finite expressions for the components of the Poisson tensor, using only the defining properties of the Darboux transformation.

Of the 28 independent components of the Poisson ten-

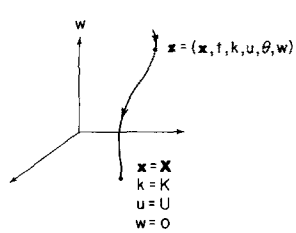


FIG. 3. A schematic illustration of a θ characteristic. The $w=0$ "plane" actually represents a six-dimensional surface in phase space.

30 October 2026 19:36:15

sor in the new coordinates, seven are built into the construction of the Darboux transformation itself, as shown by Eqs. (23a, b). Of the remaining 21 components, six will turn out to be Eq. (23c), showing that the Darboux transformation really does block-diagonalize the Poisson tensor. The other 15 will be the Poisson brackets $\{Z_i, Z_j\}$, i.e., the components of the resulting 6×6 block. We will use the action form ρ and the Lagrange tensor ω to derive the 21 (as yet) unknown components of the Poisson tensor. We now turn to a derivation of the components of the action form in the new coordinates.

We call on Eq. (19), which is valid in any coordinate system, and specialize to the new coordinates $(\mathbf{Z}, \theta, M)$. For example, the component ρ_U is given by

$$\rho_U = \left(\mathbf{D} + u\hat{b} + w\hat{c} + \frac{1}{\epsilon}\mathbf{A}\right)\cdot\frac{\partial \mathbf{x}}{\partial U} - \left(k + w\hat{c}\cdot\mathbf{D} + \frac{1}{2}D^2 + \frac{1}{\epsilon}\phi\right)\frac{\partial t}{\partial U}, \tag{25}$$

and similar equations hold for the other components, $\rho_{\mathbf{X}}$, ρ_t, ρ_K, ρ_θ, and ρ_M. Equations (4) and (12) have been used in writing Eq. (25) in the form shown, and all symbols given, such as x, are considered to be functions of the new coordinates $(\mathbf{Z}, \theta, M)$.

The time t does not change under the Darboux transformation, so the second term in Eq. (25) vanishes. The same is true for the other components, except ρ_t, for which we have $\partial t/\partial t = 1$. However, x is still an unknown function of $(\mathbf{X}, t, K, U, \theta, M)$, being precisely part of the Darboux transformation, so the derivative $\partial \mathbf{x}/\partial U$ in the first term of Eq. (25) cannot be evaluated in general. Nevertheless, if we specialize to points on the initial value surface, i.e., $M = 0$, then we have $\mathbf{x} = \mathbf{X}$ exactly, and the derivative in the first term of Eq. (25) can be evaluated.

For example, let us consider the operator $\partial/\partial U$. This operator implies that M is held fixed, so that $\partial/\partial U$ represents a directional derivative lying entirely in the initial value surface, which is $M = 0$ by construction. That is, we have

$$\left(\frac{\partial}{\partial U}\mathbf{x}(\mathbf{Z}, \theta, M)\right)_{M=0} = \frac{\partial}{\partial U}[\mathbf{x}(\mathbf{Z}, \theta, M)]_{M=0}; \tag{26}$$

hence, $\partial \mathbf{x}/\partial U = \partial \mathbf{X}/\partial U = 0$ on the initial value surface. The derivatives $\partial \mathbf{x}/\partial \mathbf{X} = \mathsf{I}$, $\partial \mathbf{x}/\partial t = 0$, $\partial \mathbf{x}/\partial K = 0$, and $\partial \mathbf{x}/\partial \theta = 0$ on the initial value surface are found in a similar way. (Since θ is not defined on the initial value surface, one may be somewhat more careful about $\partial \mathbf{x}/\partial \theta$; but, the results are the same.) The derivative $\partial \mathbf{x}/\partial M$ cannot be determined by this method, we we shall simply omit the component ρ_M in the following discussion. As we shall see, it will not be needed.

It is now trivial to write down the components of ρ on the initial value surface, which simplify greatly because of the initial conditions $(\mathbf{x}, t, k, u, \theta, w) = (\mathbf{X}, t, K, U, \theta, 0)$. The results are most conveniently expressed in terms of the modified potentials $\mathbf{A}^*$, ϕ^*, which are given by

$$\mathbf{A}^* = \mathbf{A} + \epsilon\delta\mathbf{A}, \qquad \phi^* = \phi + \epsilon\delta\phi, \tag{27}$$

where

$$\delta\mathbf{A} = U\hat{b} + \mathbf{D}, \qquad \delta\phi = \tfrac{1}{2}D^2. \tag{28}$$

These potentials are considered to be functions of $(\mathbf{X}, t, U)$, and the fields $\hat{b}$ and $\mathbf{D}$ appearing on the right-hand sides are evaluated at $(\mathbf{X}, t)$. The modified potentials have been employed previously by Morozov and Solov'ev,[17] in a rather different context. Equations (27) are not truncated power series, but are exact. For future reference we also define the modified fields $\mathbf{E}^*$ and $\mathbf{B}^*$ in the obvious way, as well as the tensor B^*_{ij} and the scalar B^*,

$$\mathbf{E}^* = \mathbf{E} + \epsilon\delta\mathbf{E} = -\nabla\phi^* - \frac{\partial \mathbf{A}^*}{\partial t}, \qquad \mathbf{B}^* = \mathbf{B} + \epsilon\delta\mathbf{B} = \nabla\times\mathbf{A}^* \tag{29}$$

$$\mathsf{B}^*_{ij} = \epsilon_{ijk}B^*_k, \qquad B^* = B + \epsilon\delta B = \hat{b}\cdot\mathbf{B}^*,$$

where ∇ means $\partial/\partial\mathbf{X}$, and where

$$\delta\mathbf{E} = -U\frac{\partial \hat{b}}{\partial t} - \frac{\partial \mathbf{D}}{\partial t} - \nabla\mathbf{D}\cdot\mathbf{D}, \qquad \delta\mathbf{B} = U\nabla\times\hat{b} + \nabla\times\mathbf{D}, \tag{30}$$

$$\delta B = U\hat{b}\cdot\nabla\times\hat{b} + \hat{b}\cdot\nabla\times\mathbf{D} = G_2(\mathbf{X}, t, U).$$

Note that B^* is not the same as $|\mathbf{B}^*|$.

Except for ρ_M, we can now summarize the components of ρ in the new coordinates, evaluated on the initial value surface

$$\rho_{\mathbf{X}} = (1/\epsilon)\mathbf{A}^*, \quad \rho_t = -K - (1/\epsilon)\phi^*, \qquad \rho_U = \rho_K = \rho_\theta = 0. \tag{31}$$

These equations are not valid for points not on the initial value surface.

D. The Lagrange and Poisson tensors in the coordinates $(\mathbf{X}, t, K, U, \theta, M)$

The seven components of ρ in Eq. (31) suffice to determine 21 components of the Lagrange tensor ω_{ij} by using Eq. (19). The calculation is trivial and is summarized in Eq. (32). In this process, one must be careful about taking derivatives, as in Eq. (26). Only 21 components are determined (instead of all 28) because the operator $\partial/\partial M$ is not used. The remaining seven components, which are $\omega_{\mathbf{Z}M}$ and $\omega_{\theta M}$, are equivalent to the seven known Poisson brackets in Eqs. (23a–b). To see this, we may order the new coordinates $(\mathbf{Z}, \theta, M)$, corresponding to the indices 1–6, 7, 8. Then, Eqs. (23a, b) can be written $J^{7k} = 0$ for $k = 1, \ldots, 6$, and $J^{78} = 1/\epsilon$. Substituting these into Eq. (18) and setting $i = 7$ gives $(1/\epsilon)\omega_{8j} = \delta^7_j$, from which the remaining seven components of ω follow.

In summary, the components of the Lagrange tensor

30 October 2026 19:36:15

in the new coordinates are

$$\omega_{X_i X_j} = (1/\epsilon) B^*_{ij}, \quad \omega_{Xt} = (1/\epsilon)\mathbf{E}^*, \qquad \omega_{XU} = -\hat{b}, \quad \omega_{tK} = 1, \quad \omega_{\theta M} = -\epsilon. \tag{32}$$

The components which are not given are zero. As constructed, these relations are valid only on the initial surface, but we shall see that they are, in fact, valid everywhere in phase space. Equations (32) may be arranged in the form of a matrix, which shows that the Lagrange tensor is 6+2 block-diagonal.

The Lagrange tensor may be inverted to find the Poisson tensor. The methods of Appendix C are helpful in this process, as is the fact that ω is block-diagonal. The results are

$$\{X_i, X_j\} = -b_{ij}/B^*, \tag{33a}$$

$$\{\mathbf{X}, K\} = b \times \mathbf{E}^*/B^*, \tag{33b}$$

$$\{\mathbf{X}, U\} = \mathbf{B}^*/B^*, \tag{33c}$$

$$\{t, K\} = -1, \tag{33d}$$

$$\{K, U\} = \mathbf{E}^* \cdot \mathbf{B}^*/(\epsilon B^*), \tag{33e}$$

$$\{\theta, M\} = 1/\epsilon. \tag{33f}$$

The brackets which are not shown are zero. The tensor b_{ij} in Eq. (33a) is defined by

$$b_{ij} = \epsilon_{ijk} b_k. \tag{34}$$

All fields appearing in Eqs. (33) are evaluated at $(\mathbf{X}, t)$.

A byproduct of the matrix inversion is the determinant of ω_{ij}, which we record here for future reference

$$\det(\omega_{ij}) = B^{*2}. \tag{35}$$

By their construction, Eqs. (33) are valid on the initial value surface, but they are also valid everywhere else in phase space. This is because the Poisson brackets shown are all constants along θ characteristics, as may be seen by the fact that they are all independent of M, the only one of the new coordinates which has a nonvanishing Poisson bracket with θ. This can also be viewed as an application of Poisson's theorem, as discussed in Ref. 1. Since the components of the Lagrange tensor are functions of the components of the Poisson tensor, these are also constants along θ characteristics, and Eqs. (32) are also valid everywhere in phase space.

It is often desirable to expand the Poisson brackets (33) in powers of ϵ. This can be done in such a way that the resulting expressions, like Eqs. (33) from which they came, are valid to all orders, if only the denominator B^* is not expanded in the remainder term. Carried to the first few terms, this procedure gives the following, alternate expressions for the Poisson brackets in Eqs. (33b, c, e):

$$\{\mathbf{X}, K\} = -\mathbf{D} + (\epsilon/B^*)\hat{b} \times (\delta\mathbf{E} + \mathbf{D} \times \delta\mathbf{B}),$$

$$\{\mathbf{X}, U\} = \hat{b} + (\epsilon/B^*)(\delta\mathbf{B})_\perp, \tag{36}$$

$$\{K, U\} = (E_\parallel/\epsilon) + (\delta\mathbf{E} + \mathbf{D} \times \delta\mathbf{B}) \cdot [\hat{b} + (\epsilon/B^*)(\delta\mathbf{B})_\perp].$$

These equations result from substituting Eqs. (29) into Eqs. (33). By using Eqs. (30), some of the terms appearing in Eqs. (36) can be written out in terms of the physical fields. We find

$$\delta\mathbf{E} + \mathbf{D} \times \delta\mathbf{B} = -U\left(\frac{\partial \hat{b}}{\partial t} - \mathbf{D} \times (\nabla \times \hat{b})\right) - \left(\mathbf{D} \cdot \nabla \mathbf{D} + \frac{\partial \mathbf{D}}{\partial t}\right),$$

$$(\delta\mathbf{B})_\perp = b \times [U\hat{b} \cdot \nabla \hat{b} - \hat{b} \times (\nabla \times \mathbf{D})],$$

$$\hat{b} \cdot (\delta\mathbf{E} + \mathbf{D} \times \delta\mathbf{B}) = U\hat{b} \cdot \nabla \hat{b} \cdot \mathbf{D} + \mathbf{D} \cdot \nabla \hat{b} \cdot \mathbf{D} + \mathbf{D} \cdot \frac{\partial \hat{b}}{\partial t} = G_3(\mathbf{X}, t, U). \tag{37}$$

E. Explicit formulas for the Darboux transformation

The Darboux transformation is explicitly evaluated by methods similar to those of Ref. 1, although the results cannot be expressed so elegantly as there. It is easiest to first find the inverse transformation, which gives z as a function of (Z, θ, M).

The θ characteristics are specified by $d\mathrm{z}/d\lambda = \{\mathrm{z}, \theta\}$, and we also have $dM/d\lambda = \{M, \theta\} = -1/\epsilon$. Since λ is a nonphysical parameter of no significance, it may be eliminated in favor of M as the independent variable of the differential equations. Formally, this gives

$$\frac{d\mathrm{z}}{dM} = \frac{\{\mathrm{z}, \theta\}}{\{M, \theta\}} = -\epsilon\{\mathrm{z}, \theta\},$$

or, explicitly in terms of the Poisson brackets in Eqs. (15),

$$\frac{d\mathbf{x}}{dM} = \epsilon \frac{\hat{a}}{w}, \quad \frac{dk}{dM} = -\frac{\epsilon}{w}(G_0 + wF_0), \qquad \frac{du}{dM} = \frac{\epsilon}{w}(G_1 + wF_1), \quad \frac{dw}{dM} = \frac{B}{w} + \frac{\epsilon}{w}(G_2 + wF_2). \tag{38}$$

The quantities F_0, G_0, etc. are defined in Eqs. (16)–(17). The equations for t and θ are omitted from Eqs. (38) since they vanish identically.

Equations (38) are solved perturbatively using the method called "parameter perturbations" by Nayfeh.[18] We write $\mathbf{x} = \mathbf{x}_0 + \epsilon\mathbf{x}_1 + \epsilon^2\mathbf{x}_2 + \ldots$, $k = k_0 + \epsilon k_1 + \ldots$, and similarly for u and w. The quantities $\mathbf{x}_0$, $\mathbf{x}_1$, k_0, k_1, etc., are considered to be functions of M; their additional dependence on X, K, and U comes from the initial conditions for the θ characteristics. These expansions are substituted into Eqs. (38) and terms are collected by order, and there results an ordered hierarchy of differential equations. For example, the first few equations are

$$\frac{d\mathbf{x}_0}{dM} = \frac{dk_0}{dM} = \frac{du_0}{dM} = 0, \qquad \frac{dw_0}{dM} = \frac{B(\mathbf{x}_0)}{w_0}, \quad \frac{d\mathbf{x}_1}{dM} = \frac{\hat{a}(\mathbf{x}_0)}{w_0}, \quad \text{etc.} \tag{39}$$

The initial conditions $(\mathbf{x}, k, u, w) = (\mathbf{X}, K, U, 0)$ when $M = 0$ are imposed on the order zero terms, and the higher order terms are taken to vanish when $M = 0$. For example, this gives $\mathbf{x}_0 = \mathbf{X}$ and $w_0 = [2MB(\mathbf{X})]^{1/2}$. Only trivial integrations are called for. The results are

30 October 2026 19:36:15

as follows:

$$\mathbf{x}=\mathbf{X}+(\epsilon/B)(2MB)^{1/2}\hat{a}+(\epsilon^2/B^2)\{-\tfrac{1}{2}(2MB)^{1/2}G_2\hat{a} -\tfrac{1}{6}(2MB)[(F_2+F_6)\hat{a}+3F_8\hat{b}+3F_9\hat{c}]\}+O(\epsilon^3), \quad (40a)$$

$$k=K-(\epsilon/B)[(2MB)^{1/2}G_0+\tfrac{1}{2}(2MB)F_0]+O(\epsilon^2), \quad (40b)$$

$$u=U+(\epsilon/B)[(2MB)^{1/2}G_1+\tfrac{1}{2}(2MB)F_1]+O(\epsilon^2), \quad (40c)$$

$$w=(2MB)^{1/2}+(\epsilon/B)[\tfrac{1}{2}(2MB)^{1/2}G_2+\tfrac{1}{3}(2MB)(F_2+F_6)] +O(\epsilon^2). \quad (40d)$$

In these expressions, all functions are evaluated at $(\mathbf{X}, t, \theta, U)$, and ∇ means $\partial/\partial\mathbf{X}$. The quantities F_0, G_0, etc. are defined in Eqs. (16)–(17).

The differential equations (38), of which Eqs. (40) are the solution, possess right-hand sides which are independent of k. Therefore, the constant of integration K appears only in an additive manner in Eqs. (40), as shown. That is, except for the first term of Eq. (40b), the right-hand sides of Eqs. (40) are independent of K to all orders. This fact will be used later.

Let us now turn to the forward Darboux transformation, which gives $\mathbf{Z}$ and M as functions of $\mathbf{z}$. There are two ways to find this transformation. One way is simply to invert Eqs. (40). The other way is to use the method of parameter perturbations again, this time with the differential equations for the θ characteristics written so that w is the independent variable. To be precise, we use the set

$$\frac{d\mathbf{z}}{dw}=\left(\frac{d\mathbf{z}}{d\lambda}\right)\left(\frac{dw}{d\lambda}\right)^{-1}=\frac{\{\mathbf{z},\theta\}}{\{w,\theta\}}, \qquad \frac{dM}{dw}=\frac{\{M,\theta\}}{\{w,\theta\}}=\frac{-1}{\epsilon\{w,\theta\}}. \quad (41)$$

As before, the variables $\mathbf{x}$, k, u, and M are ordered in powers of ϵ and the right-hand sides of Eqs. (41) are expanded. We find, for example,

$$\frac{d\mathbf{x}_0}{dw}=\frac{dk_0}{dw}=\frac{du_0}{dw}=0, \quad \frac{dM_0}{dw}=\frac{w}{B(\mathbf{x}_0)}, \qquad \frac{d\mathbf{x}_1}{dw}=\frac{\hat{a}(\mathbf{x}_0)}{B(\mathbf{x}_0)}, \text{ etc.} \quad (42)$$

We take the initial conditions for these equations to be $(\mathbf{x}, k, u, M)=(\mathbf{x}_i, k_i, u_i, M_i)$ when $w=w_i$, representing an arbitrary initial point on a θ characteristic, and impose these on the order zero terms. The higher order terms are taken to vanish when $w=w_i$. We find, for example, $\mathbf{x}_0=\mathbf{x}_i$, $M_0=M_i+(w-w_i)^2/[2B(\mathbf{x}_i)]$. Finally, when Eqs. (42) are all solved to a suitable order, we set $(\mathbf{x}, k, u, M)=(\mathbf{X}, K, U, 0)$, and $w=0$, representing a point on the initial value surface, and we set $(\mathbf{x}_i, k_i, u_i, M_i)=(\mathbf{x}, k, u, M)$, and $w_i=w$. The idea is to integrate backward along a θ characteristic to reach the initial value surface, with the roles of initial and final points reversed from those used in Eqs. (40). The results agree with the direct inversion of Eqs. (40), but the amount of algebra is less.

Thus, we have the forward Darboux transformation:

$$\mathbf{X}=\mathbf{x}-(\epsilon w/B)\hat{a}+(\epsilon^2/B^2)\{wG_2\hat{a} +(w^2/2)[(F_2-F_6)\hat{a}-F_8\hat{b}-F_9\hat{c}]\}+O(\epsilon^3), \quad (43a)$$

$$K=k+(\epsilon/B)[wG_0+(w^2/2)F_0]+O(\epsilon^2), \quad (43b)$$

$$U=u-(\epsilon/B)[wG_1+(w^2/2)F_1]+O(\epsilon^2), \quad (43c)$$

$$M=\frac{w^2}{2B}+\frac{\epsilon}{B^2}\left(-\frac{w^2}{2}G_2+\frac{w^3}{6}(F_6-2F_2)\right)+O(\epsilon^2). \quad (43d)$$

In these equations, all functions on the right-hand sides are evaluated at $(\mathbf{x}, t, \theta, u)$, and ∇ means $\partial/\partial\mathbf{x}$. The quantities F_0, G_0, etc., are defined in Eqs. (16)–(17). Arguments similar to those used in connection with Eqs. (40) show that the occurrence of the variable k on the right-hand sides of Eqs. (43) is limited to the first term of Eq. (43b).

F. The Hamiltonian in the coordinates $(\mathbf{X},t,K,U,\theta,M)$

It is a simple matter to substitute Eqs. (40) into Eq. (13) to obtain the new Hamiltonian. One finds

$$H(\mathbf{X},t,K,U,\theta,M) =MB+\tfrac{1}{2}U^2-K+(\epsilon/B)[(2MB)^{1/2}G_6 +\tfrac{1}{4}(2MB)(G_7+G_8)+\tfrac{1}{3}(2MB)^{3/2}(F_2+F_6)]+O(\epsilon^2). \quad (44)$$

All functions on the right-hand side are evaluated at $(\mathbf{X}, t, \theta, U)$, and ∇ means $\partial/\partial\mathbf{X}$. The quantities F_n and G_n are defined in Eqs. (16)–(17). The sum G_7+G_8 represents the decomposition of the term mulitplying $(2MB)$ into its averaged and oscillator parts, respectively; this will be discussed at greater length in Sec. IV. Note that $\partial H/\partial K=-1$; this holds to all orders, as may be seen by examining the K dependence of Eqs. (40).

The Hamiltonian (44) depends on the gyrophase θ in its $O(\epsilon)$ term, because we have as yet to carry out the averaging transformation. Nevertheless, this Hamiltonian may be used to obtain the well known, classic drift equations,[19] because the θ dependence of H causes corrections only at an order in ϵ which is beyond these classic drifts. To see this, let us write H in the form

$$H(\mathbf{X},t,K,U,\theta,M)=MB+\tfrac{1}{2}U^2-K +\epsilon H_1(\mathbf{X},t,U,\theta,M)+O(\epsilon^2), \quad (45)$$

and then use the Poisson bracket relations in Eqs. (33) and (36) to compute time derivatives. Let us carry the results to the highest order in ϵ which is compatible with an assumption of ignorance about the term H_1, apart from the fact that $\partial H_1/\partial K=0$. The Poisson brackets are expanded in ϵ in this process; as far as needed, this is already done in Eqs. (36).

First, we find that $\{t, H\}=dt/d\tau=1$ exactly, so that $d/d\tau=d/d\tau$. Next, we compute the time derivative of $\mathbf{X}$, which will give the drifts. Through $O(\epsilon)$, we find

$$\frac{d\mathbf{X}}{dt}=\{\mathbf{X},H\}=\mathbf{D}+U\hat{b}+\epsilon\left(\frac{1}{B}\hat{b}\times\nabla(MB) +\frac{1}{B}U(\delta B)_\perp-\frac{1}{B}\hat{b}\times(\delta E+\mathbf{D}\times\delta\mathbf{B})+\hat{b}\frac{\partial H_1}{\partial U}\right)+O(\epsilon^2). \quad (46)$$

Evidently, the parallel motion of the guiding center can

30 October 2026 19:36:15

be found only through $O(1)$, because of the term $\partial H_1/\partial U$, that is,

$$\left(\frac{d\mathbf{X}}{dt}\right)_{\parallel} = U\hat{b} + O(\epsilon) .$$

The perpendicular motion, on the other hand, can be found through $O(\epsilon)$.

$$\left(\frac{d\mathbf{X}}{dt}\right)_{\perp} = \mathbf{D} + \frac{\epsilon}{B}\hat{b}\times\left[M\nabla B + \left(U\hat{b}\cdot\nabla + \mathbf{D}\cdot\nabla + \frac{\partial}{\partial t}\right)(U\hat{b} + \mathbf{D})\right] + O(\epsilon^2) . \tag{47}$$

Equations (37) have been used to write this result in this familiar form.[19]

Mirroring effects are displayed by computing dU/dt:

$$\frac{dU}{dt} = \{U, H\} = \frac{E_{\parallel}}{\epsilon} - M\hat{b}\cdot\nabla B + \left[\left(U\hat{b}\cdot\nabla + D\cdot\nabla + \frac{\partial}{\partial t}\right)\hat{b}\right]\cdot\mathbf{D} + O(\epsilon) .$$

Note that we are assuming that $E_{\parallel} = O(\epsilon)$.

The time derivative of K can be computed similarly, but it is not needed. Since $dH/dt = 0$ exactly, we have, from the form of the Hamiltonian (45),

$$\frac{dK}{dt} = M\frac{\partial B}{\partial t} + M\frac{d\mathbf{X}}{dt}\cdot\nabla B + U\frac{dU}{dt} + O(\epsilon) . \tag{48}$$

On the phase space surface $H = 0$, K represents the kinetic energy of the particle, relative to the $\mathbf{E}\times\mathbf{B}$ drift, with the lowest order oscillations removed. Recall that the numerical value of the Hamiltonian is a nonphysical constant of integration in the extended phase space.

Finally, we have the time derivatives of θ and M,

$$\frac{d\theta}{dt} = \{\theta, H\} = \frac{B}{\epsilon} + O(1) ,$$

$$\frac{dM}{dt} = \{M, H\} = -\frac{\partial H_1}{\partial\theta} + O(\epsilon) .$$

M is the adiabatic invariant only to lowest order, so dM/dt is not zero. However, since $\partial/\partial\theta$ projects out purely oscillatory terms, M has a secular time evolution only at $O(\epsilon)$.

G. The integral invariants of Poincaré in the coordinates $(\mathbf{X}, t, K, U, \theta, M)$

The most familiar consequence of the existence of the integral invariants of Poincaré in a Hamiltonian system is Liouville's theorem. Liouville's theorem allows one to make strong assertions about the long term behavior of a Hamiltonian system of differential equations; for example, bounded attracting sets, such as limit cycles or strange attractors, cannot exist. These are important conclusions, because an arbitrary, non-Hamiltonian system of differential equations may well possess attracting sets. For these and other reasons, it has been a matter of concern in the past that the familiar guiding center equations of motion do not conserve phase space volume in the obvious way, i.e., $\nabla\cdot\dot{\mathbf{X}} + \partial\dot{U}/\partial U \neq 0$, even to lowest order. In this subsection we will present a set of guiding center equations which exactly preserve all of the integral invariants of Poincaré to all orders.

Of the N integral invariants of Poincaré in a system of N degrees of freedom, usually only the first and the last are of practical importance. The last Poincaré invariant yields Liouville's theorem; we will discuss it first. The discussion will be clearer if we restrict our consideration to static magnetic fields with $\mathbf{E} = 0$; we will temporarily deal with a six-dimensional phase space with coordinates $(\mathbf{X}, U, \theta, M)$ (see Appendix A). Later we will generalize the results to time-dependent fields.

Let us change notation somewhat, and write $\mathbf{z}' = (\mathbf{q}, \mathbf{p})$ for the oroginal canonical coordinates of Sec. II A, and $\mathbf{z} = (\mathbf{X}, U, \theta, M)$ for the guiding center variables of this section. Let us write ω'_{ij} for the Lagrange tensor in the coordinates $\mathbf{z}'$, and ω_{ij} for the same tensor in the coordinates $\mathbf{z}$. Then, the transformation law for covariant tensors takes the form

$$\omega_{ij} = \frac{\partial z'_k}{\partial z_i}\omega'_{kl}\frac{\partial z'_l}{\partial z_j} . \tag{49}$$

But, ω'_{kl} is the simple, constant matrix of Lagrange brackets in the coordinates $\mathbf{z}' = (\mathbf{q}, \mathbf{p})$; it has components $\omega'_{pq} = -\omega'_{qp} = 1$, with all others being zero. Hence, $\det(\omega'_{ij}) = 1$ and Eq. (49) implies

$$\det\left|\frac{\partial\mathbf{z}'}{\partial\mathbf{z}}\right| = \det(\omega_{ij})^{1/2} , \tag{50}$$

or, by using Eq. (35),

$$\det[\partial(\mathbf{q}, \mathbf{p})/\partial(\mathbf{X}, U, \theta, M)] = B^* , \tag{51}$$

exactly. Hence, we have

$$d^3\mathbf{q}\, d^3\mathbf{p} = B^* d^3\mathbf{X} dU\, d\theta\, dM . \tag{52}$$

This relation is of obvious importance in any Vlasov theory of a plasma which uses the coordinates $(\mathbf{X}, U, \theta, M)$. As we shall see, Eq. (52) is valid even for time-dependent fields.

Equation (52) suggests that B^* represents, as it were, the density of the phase space fluid. Hence, we expect Liouville's theorem to have the form

$$\frac{\partial}{\partial\mathbf{z}}\cdot(B^*\dot{\mathbf{z}}) = \frac{\partial}{\partial\mathbf{z}}\cdot(B^*\{\mathbf{z}, H\}) = 0 , \tag{53}$$

for any Hamiltonian H. Indeed, by using Eqs. (33), this can easily be shown to hold to all orders; the proof reduces to $\nabla\cdot\mathbf{B}^* = 0$. We see that in the coordinates $\mathbf{z} = (\mathbf{X}, U, \theta, M)$, the phase fluid is compressible, but it has a mass conservation law. Note that if $\partial H/\partial\theta = 0$, then $\dot{\theta}$ is independent of θ and $\dot{M} = 0$, so Eq. (53) reduces to

$$\frac{\partial}{\partial\mathbf{X}}\cdot(B^*\dot{\mathbf{X}}) + \frac{\partial}{\partial U}(B^*\dot{U}) = 0 . \tag{54}$$

In practice, the use of a truncated power series for the Hamiltonian is unavoidable. However, as long as the exact Poisson brackets given in Eqs. (33) and (36) are used, the validity of Liouville's theorem is not affected. For example, let us take only the lowest order

30 October 2026 19:36:15

term in the Hamiltonian, $H=MB+\frac{1}{2}U^2$, and write out the equations of motion which result from using the exact Poisson brackets. We find

$$\frac{d\mathbf{X}}{dt}=\hat{b}U+\frac{\epsilon}{B^*}\hat{b}\times(M\nabla B+U^2\hat{b}\cdot\nabla\hat{b}), \tag{55a}$$

$$\frac{dU}{dt}=-\left(\hat{b}+\frac{\epsilon}{B^*}U\hat{b}\times(\hat{b}\cdot\nabla\hat{b})\right)\cdot(M\nabla B). \tag{55b}$$

These equations differ from the usual guiding center equations, which we display here for comparison.

$$\frac{d\mathbf{X}}{dt}=\hat{b}U+\frac{\epsilon}{B}\hat{b}\times(M\nabla B+U^2\hat{b}\cdot\nabla\hat{b}), \tag{56a}$$

$$\frac{dU}{dt}=-M\hat{b}\cdot\nabla B. \tag{56b}$$

On the basis of ϵ ordering alone, neither (55) nor (56) can be considered more accurate, because of the neglected terms in the Hamiltonian. We suggest, however, that in terms of the difference between the trajectory of the guiding center and that of the particle, Eqs. (55) will have better behaved error characteristics, because they satisfy Liouville's theorem. Some numerical experiments on this question would be interesting.

An issue related to this is the conservation of energy. Equations (56) do not conserve the energy $MB+\frac{1}{2}U^2$, as may easily be verified by direct substitution. Of course, this energy differs from the true energy of the particle by higher order terms, but it cannot be considered a positive feature of Eqs. (56) that they do not possess any simple constant of integration, even an approximate energy. Equations (55), on the other hand, conserve the truncated energy exactly.

Let us generalize these results to time-dependent fields. We now return to the eight-dimensional phase space, with coordinates $(\mathbf{q}, \mathbf{p}, h, t)$ or $(\mathbf{X}, U, \theta, M, K, t)$. Proceeding as before, we find

$$\det\left(\frac{\partial(\mathbf{q},\mathbf{p},h,t)}{\partial(\mathbf{X},U,\theta,M,K,t)}\right)=B^*. \tag{57}$$

The volume element in the extended phase space is a rather unphysical object, because of the unphysical use of the variables h and t or K and t. However, Eq. (57) can be reduced to obtain the Jacobian of $(\mathbf{q},\mathbf{p})\rightarrow(\mathbf{X},U,\theta,M)$, as in Eq. (51). To do this, we observe from Eqs. (4), (12), and (40) that

$$\frac{\partial\mathbf{q}}{\partial K}=\frac{\partial\mathbf{p}}{\partial K}=\frac{\partial t}{\partial\mathbf{X}}=\frac{\partial t}{\partial U}=\frac{\partial t}{\partial M}=\frac{\partial t}{\partial\theta}=\frac{\partial t}{\partial K}=0,\quad \frac{\partial h}{\partial K}=\frac{\partial t}{\partial t}=1. \tag{58}$$

By using these relations and expanding by cofactors, the determinant in Eq. (57) can be reduced by two orders. One finds that Eq. (52) is valid even in the time-dependent case, although B^* has extra terms; see Eqs. (27)–(29).

Liouville's theorem, in the form of Eq. (53), goes directly over to the time-dependent case, with z now being the coordinate vector $(\mathbf{X}, U, \theta, M, K, t)$. For any Hamiltonian satisfying $\partial H/\partial\theta=0$ and $\partial H/\partial K=-1$, such as the lowest order truncation of Eq. (45), several terms vanish, and we have

$$\frac{\partial B^*}{\partial t}+\frac{\partial}{\partial\mathbf{X}}\cdot(B^*\dot{\mathbf{X}})+\frac{\partial}{\partial U}(B^*\dot{U})=0.$$

This is the obvious generalization of Liouville's theorem to time-dependent fields; the density of the phase space fluid B^* is now time dependent.

Let us now briefly examine the first integral invariant of Poincaré, which is connected in practice with the construction of action variables. This may be regarded either as the integral of ω over a two-dimensional surface in phase space, or as the line integral of the action form ρ around its boundary.

The action form ρ on the initial value surface is shown in Eqs. (31), but it does not help us because we require an expression valid everywhere in phase space. Such an expression may be found by working backward from ω to ρ, using Eqs. (32) and (19). Simple inspection gives

$$\rho_{\mathbf{X}}=(1/\epsilon)\mathbf{A}^*,\quad \rho_t=-K-(1/\epsilon)\phi^*,\quad \rho_K=\rho_U=\rho_M=0,\ \rho_\theta=\epsilon M. \tag{59}$$

It is trivial to verify Eq. (19) using this result for ρ. Equation (59) does not reduce to Eq. (31)) when $M=0$, because the action form given there and the ρ given here differ by the total differential of a scalar W[see Eq. (21)].

From Eq. (59) we have the action integral associated with a closed loop in phase space

$$J=\frac{1}{2\pi}\oint\left[\frac{1}{\epsilon}\mathbf{A}^*\cdot d\mathbf{X}-\left(K+\frac{1}{\epsilon}\phi^*\right)+\epsilon M\,d\theta\right]. \tag{60}$$

This integral enters prominently into Kruskal's theory of nearly periodic systems,[20] and it has applications in adiabatic theory.[21]

IV. THE AVERAGING TRANSFORMATION

The averaging transformation is a symplectic transformation which removes the rapid oscillations from the guiding center variables. It has the form $(\mathbf{X}, t, K, U, \theta, M)\rightarrow(\overline{\mathbf{X}}, t, \overline{K}, \overline{U}, \overline{\theta}, \overline{M})$, in which the overbars indicate that the new variables are averaged, i.e., free of rapid oscillations. Because the averaging transformation is symplectic, all of the exact results derived in Sec. III for the old variables are also valid in the new variables. This property of symplectic transformations, i.e., the fact that the functional form of the Poisson and Lagrange tensors is invariant under them, was discussed in detail in Ref. 1.

A. Lie transforms

Except for a few minor complications, the Lie transform procedure is quite similar to that used in Ref. 1. We denote the generators of the symplectic transformation by $g_1, g_1, \ldots$, the associated Lie operators by $L_1, L_2, \ldots$, and the elementary symplectic transformations by $T_1, T_2, \ldots$. These are related to one another by

$$L_n=\epsilon\{g_n,\ \},\quad T_n=\exp[-(\epsilon^n/n)L_n]. \tag{61}$$

30 October 2026 19:36:15

We restrict our consideration to generators which satisfy $\{g_n, t\}=0$, so that the time t does not change under the averaging transformation. By Eqs. (33), this is equivalent to $\partial g_n/\partial K=0$.

Since the operators L_n depend on ϵ, we expand them in powers of ϵ:

$$L_n=\sum_{m=0}^{\infty}\epsilon^m L_{nm}\,, \tag{62}$$

where the L_{nm}, like the L_n, are differential operators. Using Eqs. (36) and (37), we may write out L_{n0} and L_{n1}, assuming $\partial g_n/\partial K=0$ and $E_{\parallel}=O(\epsilon)$. We find

$$L_{n0}=\frac{\partial g_n}{\partial\theta}\frac{\partial}{\partial M}-\frac{\partial g_n}{\partial M}\frac{\partial}{\partial\theta}\,,$$

$$L_{n1}=\hat{b}\cdot\Delta g_n\frac{\partial}{\partial U}-\frac{\partial g_n}{\partial U}\hat{b}\cdot\nabla$$

$$-\left[\left(\frac{1}{\epsilon}E_{\parallel}+G_3\right)\frac{\partial g_n}{\partial U}+\mathbf{D}\cdot\nabla g_n+\frac{\partial g_n}{\partial t}\right]\frac{\partial}{\partial K}\,, \tag{63}$$

where G_3 is evaluated at $(\mathbf{X}, t, U)$ [see Eqs. (17)].

The operators T_n are multiplied together to form the overall symplectic transformation $T=\cdots T_3T_2T_1$. Each T_n is expanded in a power series in ϵ by using Eqs. (61)–(62), and these power series are multiplied together to obtain a series for T. The result is

$$T=I-\epsilon L_{10}+\frac{\epsilon^2}{2}(-L_{20}+L_{10}^2-2L_{11})+O(\epsilon^3)\,. \tag{64}$$

It is now convenient to write $\mathbf{z}=(\mathbf{X}, t, K, U, \theta, M)$ for the old variables, and $\bar{\mathbf{z}}=(\bar{\mathbf{X}}, t, \bar{K}, \bar{U}, \bar{\theta}, \bar{M})$ for the new. Then, we have $\bar{\mathbf{z}}=T\mathbf{z}$, and $T\bar{H}=H$, which connects the old Hamiltonian H to the new Hamiltonian $\bar{H}$. This latter relation is equivalent to $\bar{H}(\bar{\mathbf{z}})=H(\mathbf{z})$.

The old and new Hamiltonians are ordered in ϵ, i.e.,

$$H=\sum_{n=0}^{\infty}\epsilon^n H_n\,,\quad \bar{H}=\sum_{n=0}^{\infty}\epsilon^n\bar{H}_n\,. \tag{65}$$

By using Eqs. (64)–(65), the relation $T\bar{H}=H$ is multiplied out, and it yields an ordered hierarchy of equations connecting the g_n and $\bar{H}_n$. The first two are

$$0=\bar{H}_0-H_0\,,$$

$$L_{10}H_0=\bar{H}_1-H_1\,. \tag{66}$$

A common conceptual problem with Lie transforms concerns the distinction between functions and the values of functions. The operators L_{nm} act on functions, not values; hence, the independent arguments are dummies, and may be either $\mathbf{z}$ or $\bar{\mathbf{z}}$.

B. The averaged Hamiltonian

Let us specialize the Lie transform formulas to the Hamiltonian given in Eq. (44). As usual with Lie transforms, we demand that the $\bar{H}_n$ be purely averaged (i.e., independent of θ), and that the g_n be free of secular terms (i.e., $\partial g_n/\partial\theta$ is pure oscillatory). At order zero we have, trivially,

$$\bar{H}_0(\bar{\mathbf{X}}, t, \bar{K}, \bar{U}, \bar{M})=\bar{M}B(\bar{\mathbf{X}}, t)+\tfrac{1}{2}\bar{U}^2-\bar{K}\,. \tag{67}$$

Next, at order one we have

$$B\frac{\partial g_n}{\partial\theta}=-\mathrm{osc}(H_1)\,, \tag{68}$$

$$\bar{H}_1=\mathrm{avg}(H_1)\,, \tag{69}$$

where avg represents an averaging over θ and osc is the complementary operator.

It may already be seen, in Eqs. (67) and (69), that $\partial\bar{H}/\partial\bar{K}=-1$. This result holds to all orders, as did the analogous result for H and K, and it is easily proved by calling on the assumption $\partial g_n/\partial K=0$.

It was mentioned previously that the term G_7 in Eq. (44) is purely averaged, and G_8 is purely oscillatory. As we shall see, the terms G_6, F_2, and F_6 in Eq. (44) are also purely oscillatory. Granting this for the moment, we have

$$\mathrm{avg}(H_1)=(M/2)G_7\,, \tag{70}$$

$$\mathrm{osc}(H_1)=(1/B)[(2MB)^{1/2}G_6+\tfrac{1}{4}(2MB)G_8$$
$$+\tfrac{1}{3}(2MB)^{3/2}(F_2+F_6)]\,. \tag{71}$$

In order to justify these result and show how the avg and osc projections are carried out, we turn to a consideration of the θ dependence of the terms in question.

H_1 depends on θ only through the quantities F_n and G_n [see Eqs. (16)–(17)], and these in turn are polynomial expressions in the vectors $\hat{a}$ and $\hat{c}$. These vectors are linear in $\sin\theta$ and $\cos\theta$ [see Eq. (11)], so a polynomial expression of degree n in $\hat{a}$ or $\hat{c}$ will in general contain the harmonics n, $n-2$, $n-4$, etc., in θ. Therefore, only an even polynomial in $\hat{a}$ or $\hat{c}$ can give rise to an averaged part. But, it is easy to see that the factors in Eq. (44) which multiply $(2MB)^{n/2}$ are polynomials of degree n in $\hat{a}$ and $\hat{c}$, so only the one multiplying $(2MB)$ can give an averaged part, as indicated by Eq. (70).

Let us write out the quantity G_7+G_8 and analyze its Fourier decomposition. We have

$$G_7+G_8=2\left(\hat{c}\cdot\nabla\mathbf{D}\cdot\hat{a}+\mathbf{D}\cdot\nabla\hat{c}\cdot\hat{a}+\hat{a}\cdot\frac{\partial\hat{c}}{\partial t}\right)$$
$$+2U(\hat{c}\cdot\nabla\hat{b}\cdot\hat{a}+\hat{b}\cdot\nabla\hat{c}\cdot\hat{a})\,. \tag{72}$$

The vector $\nabla\hat{c}\cdot\hat{a}$ and the scalar $\hat{a}\cdot\partial\hat{c}/\partial t$ are independent of θ, i.e., purely averaged, even though they are quadratic in $\hat{a}$ and $\hat{c}$. To emphasize this, we write $\mathbf{R}$ and S for these quantities, respectively. The geometrical meaning of $\mathbf{R}$ and S is discussed in Sec. IV E. For now, we simply show that they are indeed independent of θ, by substituting from Eq. (11). We find

$$\mathbf{R}=\nabla\hat{c}\cdot\hat{a}=\nabla\hat{\tau}_1\cdot\hat{\tau}_2-\nabla\psi\,, \tag{73a}$$

$$S=\hat{a}\cdot\frac{\partial\hat{c}}{\partial t}=\hat{\tau}_2\cdot\frac{\partial\hat{\tau}_1}{\partial t}-\frac{\partial\psi}{\partial t}\,. \tag{73b}$$

Therefore, we can write Eq. (72) as

$$G_7+G_8=2\hat{a}\hat{c}:(\nabla\mathbf{D}+U\nabla\hat{b})+2(\mathbf{D}+U\hat{b})\cdot\mathbf{R}+2S\,, \tag{74}$$

in which it is seen that all θ dependence is contained in the tensor $\hat{a}\hat{c}$. {We adopt the convention that a double contraction takes place on innermost, adjacent operands first; thus, for example, $\hat{a}\hat{c}:\nabla\mathbf{D}=\hat{a}\cdot[(\hat{c}\cdot\nabla)\mathbf{D}]=\hat{c}\cdot\nabla\mathbf{D}\cdot\hat{a}$.}

30 October 2026 19:36:15

We now introduce the following four tensors:

$$\mathsf{P}_0=\hat{a}\hat{a}+\hat{c}\hat{c}\,,\quad \mathsf{Q}_0=\hat{a}\hat{c}+\hat{c}\hat{a}\,,$$
$$\mathsf{P}_1=\hat{a}\hat{c}-\hat{c}\hat{a}\,,\quad \mathsf{Q}_1=\hat{a}\hat{a}-\hat{c}\hat{c}\,. \tag{75}$$

If T represents any one of these tensors, then T satisfies a differential equation of the form $\partial^2\mathsf{T}/\partial\theta^2=-n^2\mathsf{T}$, which shows that T is purely of the n'th harmonic in θ (see also Appendix B). For the P's, $n=0$; these are purely averaged. For the Q's, $n=2$; these are purely oscillatory, of the second harmonic. Since $\hat{a}\hat{c}=\frac{1}{2}(\mathsf{P}_1+\mathsf{Q}_0)$, the final decomposition of Eq. (74) into averaged and oscillatory parts is immediate, and it is easily verified that these parts are G_7 and G_8, respectively, as claimed. That is,

$$\text{avg}(G_7+G_8)=\mathsf{P}_1:(\nabla\mathbf{D}+U\nabla\hat{b})+2(\mathbf{D}+U\hat{b})\cdot\mathbf{R}+S=G_7\,, \tag{76a}$$

$$\text{osc}(G_7+G_8)=\mathsf{Q}_0:(\nabla\mathbf{D}+U\nabla\hat{b})=G_8\,. \tag{76b}$$

These expressions are alternate forms for G_7 and G_8, in addition to those given in Eq. (17).

Collecting these results together we have the new Hamiltonian

$$\overline{H}(\overline{\mathbf{X}},t,\overline{K},\overline{U},\overline{M})=\overline{M}B+\tfrac{1}{2}\overline{U}^2-\overline{K}+(\epsilon/2)\overline{M}G_7(\overline{\mathbf{X}},t,\overline{U})+O(\epsilon^2)\,. \tag{77}$$

We will return to this Hamiltonian when we discuss the drift equations.

C. The averaging transformation

The averaging transformation is determined by applying the operator T in Eq. (64) to the old coordinates $(\mathbf{X},t,K,U,\theta,M)$. Assuming we know g_1, but not g_2, and using Eqs. (63), we can write

$$\overline{\mathbf{X}}=\mathbf{X}+\epsilon^2\hat{b}\,\frac{\partial g_1}{\partial U}+O(\epsilon^3)\,, \tag{78a}$$

$$\overline{U}=U+O(\epsilon^2)\,, \tag{78b}$$

$$\overline{K}=K+O(\epsilon^2)\,, \tag{78c}$$

$$\overline{\theta}=\theta+\epsilon\,\frac{\partial g_1}{\partial M}+O(\epsilon^2)\,, \tag{78d}$$

$$\overline{M}=M-\epsilon\,\frac{\partial g_1}{\partial\theta}+O(\epsilon^2)\,. \tag{78e}$$

To find g_1 itself, we combine Eqs. (68) and (71) to obtain

$$\frac{\partial g_1}{\partial\theta}=-\frac{1}{B^2}[(2MB)^{1/2}G_6+\tfrac{1}{4}(2MB)G_8+\tfrac{1}{3}(2MB)^{3/2}(F_2+F_6)]\,.$$

By using the formulas of Appendix B and Eqs. (16), (17), and (76b), the integration of G_6, G_8, and F_6 is immediate; and so is that of F_2, once we realize that $F_2=\hat{c}\cdot\mathbf{R}$, so that F_2 is purely of the first harmonic in θ. We find

$$g_1=(1/B^2)[(2MB)^{1/2}G_9+\tfrac{1}{8}(2MB)G_{10}+\tfrac{1}{3}(2MB)^{3/2}(F_7-F_9)]\,, \tag{79}$$

where the symbols G_n, F_n are defined in Eqs. (16)–(17). In analogy to Eq. (76b), we have an alternate form for G_{10}

$$G_{10}=-\mathsf{Q}_1:(\nabla\mathbf{D}+U\nabla\hat{b})\,. \tag{80}$$

It is now straightforward to substitute Eq. (79) into Eq. (78) to obtain the averaging transformation explicitly. We find

$$\overline{\mathbf{X}}=\mathbf{X}+(\epsilon^2/B^2)\hat{b}[(2MB)^{1/2}G_{11}+\tfrac{1}{8}(2MB)(F_5-F_8)]+O(\epsilon^3)\,, \tag{81a}$$

$$\overline{K}=K+O(\epsilon^2)\,, \tag{81b}$$

$$\overline{U}=U+O(\epsilon^2)\,, \tag{81c}$$

$$\overline{\theta}=\theta+(\epsilon/B)[(2MB)^{-1/2}G_9+\tfrac{1}{4}G_{10}+(2MB)^{1/2}(F_7-F_9)]+O(\epsilon^2)\,, \tag{81d}$$

$$\overline{M}=M+(\epsilon/B^2)[(2MB)^{1/2}G_6+\tfrac{1}{4}(2MB)G_8+\tfrac{1}{3}(2MB)^{3/2}(F_2+F_6)]+O(\epsilon^2)\,. \tag{81e}$$

In these formulas the functions on the right-hand sides, including the quantities F_n and G_n, are evaluated at $(\mathbf{X},t,\theta,U)$, and ∇ means $\partial/\partial\mathbf{X}$.

The inverse of this transformation need not be written down, because to the order given, it may be obtained by simply swapping the plain and overbarred variables and changing the sign of the correction terms.

Of greater interest than the transformation (81) is its composition with the Darboux transformation, which will give the averaged variables as functions of the original physical particle variables. Combining Eqs. (81) and (43), we find

$$\overline{\mathbf{X}}=\mathbf{x}-(\epsilon w/B)\hat{a}+(\epsilon^2/B^2)\{w(G_2\hat{a}+G_{11}\hat{b})+(w^2/8)[4(F_2-F_6)\hat{a}+(F_5-5F_8)\hat{b}-4F_9\hat{c}]\}+O(\epsilon^3)\,, \tag{82a}$$

$$\overline{K}=k+(\epsilon/B)(wG_0+(w^2/2)F_0)+O(\epsilon^2)\,, \tag{82b}$$

$$\overline{U}=u-(\epsilon/B)(wG_1+(w^2/2)F_1)+O(\epsilon^2)\,, \tag{82c}$$

$$\overline{\theta}=\theta+(\epsilon/B)[(1/w)G_9+\tfrac{1}{4}G_{10}+w(F_7-F_9)]+O(\epsilon^2)\,, \tag{82d}$$

$$\overline{M}=w^2/2B+(\epsilon/B^2)[wG_6+(w^2/4)(G_8-2G_2)+(w^3/2)F_6]+O(\epsilon^2)\,. \tag{82e}$$

All functions on the right-hand sides of these equations, including the F_n and G_n, are evaluated at $(\mathbf{x},t,\theta,u)$, and ∇ means $\partial/\partial\mathbf{x}$. [See Eqs. (16)–(17) for the definitions of the F_n and G_n.]

The inverse of this transformation gives the physical particle variables as functions of the averaged variables. It is the composition of Eq. (40) with the inverse of Eq. (81), and it is given by

$$\mathbf{x}=\overline{\mathbf{X}}+(\epsilon/B)(2\overline{M}B)^{1/2}\hat{a}+(\epsilon^2/B^2)\{-(G_8\hat{a}+G_9\hat{c})-\tfrac{1}{4}(2\overline{M}B)^{1/2}[(G_8+2G_2)\hat{a}+4G_{11}\hat{b}+G_{10}\hat{c}]-\tfrac{1}{8}(2\overline{M}B)[4(F_2+F_6)\hat{a}+(F_5+3F_8)\hat{b}+4(2F_7-F_9)\hat{c}]\}+O(\epsilon^3)\,, \tag{83a}$$

$$k=\overline{K}-(\epsilon/B)[(2\overline{M}B)^{1/2}G_0+\tfrac{1}{2}(2\overline{M}B)F_0]+O(\epsilon^2)\,, \tag{83b}$$

$$u=\overline{U}+(\epsilon/B)[(2\overline{M}B)^{1/2}G_1+\tfrac{1}{2}(2\overline{M}B)F_1]+O(\epsilon^2)\,, \tag{83c}$$

30 October 2026 19:36:15

$$\theta = \bar{\theta} - (\epsilon/B)[(2\bar{M}B)^{-1/2}G_9 + \tfrac{1}{4}G_{10} + (2\bar{M}B)^{1/2}(F_7 - F_9)] + O(\epsilon^2), \tag{83d}$$

$$w = (2\bar{M}B)^{1/2} + (\epsilon/B)[-G_8 + \tfrac{1}{4}(2\bar{M}B)^{1/2}(2G_2 - G_8)] + O(\epsilon^2). \tag{83e}$$

All functions on the right-hand sides of these equations, including the F_n and G_n, are evaluated at $(\bar{\mathbf{X}}, t, \bar{\theta}, \bar{U})$, and ∇ means $\partial/\partial\bar{\mathbf{X}}$.

Of all these results, special interest is attached to Eq. (82e), because this equation gives the adiabatic invariant $\bar{M}$ which is a formal constant of the motion. The $O(\epsilon)$ term of this series was first worked out by Kruskal,[19] and the expression given here agrees with his result. For the case of time-independent fields with $\mathbf{E} = 0$, a recursion relation has been derived for the terms of this series by Hastie, Taylor, and Haas.[22] These authors have used their recursion relation to determine the $O(\epsilon^2)$ term. For the special case of axisymmetric fields, the same term has been derived by Gardner,[11] using Hamiltonian methods.

D. A choice for ψ; the equations of motion

Let us return to the Hamiltonian (77). The $O(\epsilon)$ term of this Hamiltonian depends on the field ψ, as may be seen from Eqs. (76a) and (73). Since ψ has been arbitrary up to this point, we ask whether an appropriate choice for ψ will simplify the Hamiltonian.

The answer is yes. There are different possibilities, but one of the most convenient is

$$\psi(\mathbf{x}, t) = \int^{\mathbf{x}} \left(\frac{1}{2}\nabla\times\hat{b} + \nabla\hat{\tau}_1\cdot\hat{\tau}_2\right)\cdot d\mathbf{x}', \tag{84}$$

where the integration is taken along a field line, at fixed time, relative to an arbitrary initial value surface. We will give a geometrical interpretation of this choice for ψ in Sec. IV E.

The virtue of this choice for ψ is that it gives

$$\hat{b}\cdot\mathbf{R} = -\tfrac{1}{2}\hat{b}\cdot(\nabla\times\hat{b}), \tag{85}$$

which simplifies G_7

$$G_7 = \hat{b}\cdot(\nabla\times\mathbf{D}) + 2\mathbf{D}\cdot\mathbf{R} + 2S. \tag{86}$$

Note in particular that $\partial G_7/\partial U = 0$.

These relations simplify the equations of motion. Let us use the Poisson brackets of Eqs. (33) and (36), the Hamiltonian (77), and the relation (86) to write down the equations of motion. We find

$$\left(\frac{d\bar{\mathbf{X}}}{dt}\right)_{\parallel} = \hat{b}\bar{U}, \tag{87a}$$

$$\left(\frac{d\bar{\mathbf{X}}}{dt}\right)_{\perp} = \mathbf{D} + \frac{\epsilon}{B^*}\hat{b}\times\left[\bar{M}\nabla B + \left(\bar{U}\hat{b}\cdot\nabla + \mathbf{D}\cdot\nabla + \frac{\partial}{\partial t}\right)(\bar{U}\hat{b} + \mathbf{D})\right] + \frac{\epsilon^2\bar{M}}{2B^*}\hat{b}\times\nabla G_7, \tag{87b}$$

$$\frac{d\bar{U}}{dt} = \frac{E_{\parallel}}{\epsilon} + \left[\left(\bar{U}\hat{b}\cdot\nabla + \mathbf{D}\cdot\nabla + \frac{\partial}{\partial t}\right)\hat{b}\right]\cdot\mathbf{D} - \bar{M}[\hat{b} + (\epsilon/B^*)(\delta\mathbf{B})_{\perp}]\cdot[\nabla B + (\epsilon/2)\nabla G_7] + (\epsilon/B^*)(\delta\mathbf{B})_{\perp}\cdot(\delta\mathbf{E} + \mathbf{D}\times\delta\mathbf{B}), \tag{87c}$$

$$\frac{d\bar{\theta}}{dt} = \frac{B}{\epsilon} + \frac{1}{2}G_7. \tag{87d}$$

[See Eqs. (17), (28), and (37) for definitions and relations concerning G_7, $\delta\mathbf{E}$, and $\delta\mathbf{B}$.]

The exact Poisson bracket relations (33) have been used in deriving Eqs. (87), so these equations preserve all the Poincaré invariants, and, in the time-independent case, the energy as well. If it is only desired that they should be accurate to a consistent order in ϵ, then the denominators in B^* can be expanded out, and Eqs. (87b) and (87c) can be truncated after $O(\epsilon^2)$ and and $O(\epsilon)$, respectively.

In the special case of purely magnetostatic fields, the effect of Eq. (84) is even more striking, for in that case we have $G_7 = 0$, and

$$\bar{H}(\bar{\mathbf{X}}, \bar{U}, \bar{M}) = \bar{M}B + \tfrac{1}{2}\bar{U}^2 + O(\epsilon^2). \tag{88}$$

In this case the equations of motion are the same as given earlier in Eqs. (55), but valid to one higher order.

Because of the relative simplicity of Eqs. (87), the variable set $(\bar{\mathbf{X}}, t, \bar{K}, \bar{U}, \bar{\theta}, \bar{M})$, with ψ given by Eq. (84), is convenient for algebraic calculations, although Eq. (85) must be carried along as an auxiliary relation. However, Eq. (84) is not practical for numerical integration. For numerical work, the equations of Appendix D are recommended.

Although Eq. (84) simplifies the Hamiltonian, the latter still depends on the original choice of perpendicular unit vectors, as do the variables $\bar{\mathbf{X}}$ and $\bar{K}$ and the equations of motion. We will return to this issue in Sec. V.

E. Geometrical interpretation of the quantities R and S

We have seen that the quantities $\mathbf{R}$ and S are dependent on the choice made for $\psi(\mathbf{x}, t)$, or, equivalently, on the original selection of the perpendicular unit vectors $\hat{\tau}_1$ and $\hat{\tau}_2$. These quantities appear in the Hamiltonian of Eq. (77) and in the definitions of the guiding center variables in terms of the physical particle variables, Eqs. (82). Here, we will give a geometrical interpretation to these quantities and show that their appearance at least somewhere in the theory is unavoidable.

The magnetic field and the perpendicular plane defined by it are physical constructs, but the perpendicular plane itself only determines the pair of perpendicular unit vectors $\hat{\tau}_1$ and $\hat{\tau}_2$ to within a rotation about $\hat{b}$. Therefore, a choice of perpendicular unit vectors introduces a certain nonphysical element into the theory, an element which in this work is embodied in the field ψ. Essentially, ψ represents a rotation angle between two possible choices of perpendicular unit vectors.

The meaning of the quantities $\mathbf{R}$ and S is best seen by examining the time evolution of the gyrophase. Let us revert to an arbitrary ψ, and use the Hamiltonian (77) to compute $d\bar{\theta}/dt$. We find

$$\frac{d\bar{\theta}}{dt} = \frac{B}{\epsilon} + \frac{1}{2}\hat{b}\cdot\nabla\times(\bar{U}\hat{b} + \mathbf{D}) + (\bar{U}\hat{b} + \mathbf{D})\cdot\mathbf{R} + S. \tag{89}$$

30 October 2026 19:36:15

The terms containing R and S are essentially geometrical in origin, and have to do with the constraints placed on $\hat{\tau}_1$ and $\hat{\tau}_2$ as a function of $(\mathbf{x}, t)$, due to the requirement that they be perpendicular to $\hat{b}$. To see this, consider two neighboring points x and $\mathbf{x}+d\mathbf{x}$, and the perpendicular unit vectors at the two points. The vectors $\hat{\tau}_1$ and $\hat{\tau}_2$ and the plane they span will be rotated at $\mathbf{x}+d\mathbf{x}$ relative to their counterparts at x. An angle of rotation of the vectors about $\hat{b}$ can be defined by projecting $\hat{\tau}_1$ and $\hat{\tau}_2$ at $\mathbf{x}+d\mathbf{x}$ back onto the perpendicular plane defined at x, and then comparing with the values of $\hat{\tau}_1$ and $\hat{\tau}_2$ at x. Indeed, if we let $d\alpha$ represent the angle in question, then we have $d\alpha = d\mathbf{x}\cdot\nabla\hat{\tau}_1\cdot\hat{\tau}_2$. Likewise, if we stand in one place and wait for time dt, $\hat{\tau}_1$ and $\hat{\tau}_2$ will rotate by an angle $d\alpha = dt\hat{\tau}_2\cdot\partial\hat{\tau}_1/\partial t$. Altogether, a point moving on a trajectory $\mathbf{x}=\mathbf{x}(t)$ would see the perpendicular unit vectors rotate about $\hat{b}$ at an angular frequency

$$\frac{d\alpha}{dt} = \frac{d\mathbf{x}}{dt}\cdot\nabla\hat{\tau}_1\cdot\hat{\tau}_2 + \hat{\tau}_2\cdot\frac{\partial\hat{\tau}_1}{\partial t}\,. \tag{90}$$

Combining this with Eqs. (9d) and (73), we see that the last two terms of Eq. (89) represent the twisting of $\hat{\tau}_1$ and $\hat{\tau}_2$ (and the gradient of ψ) which the particle sees as it drifts with the (lowest order) velocity $\overline{U}\hat{b}+\mathbf{D}$. It is as if the particle carries with it a two-dimensional inertial frame, which it transports without any rotation about $\hat{b}$ as it drifts; the last two terms of Eq. (89) represent the difference between the inertial frame and the $(\hat{\tau}_1, \hat{\tau}_2)$ frame.

It is natural to ask whether it is possible to define $\hat{\tau}_1$ and $\hat{\tau}_2$ so that they do not rotate about $\hat{b}$ as the particle moves from point to point. If this were possible, then we could set $\psi=0$, and the last two terms of Eq. (89) would vanish. Let us imagine that we define $\hat{\tau}_1$ and $\hat{\tau}_2$ at a single point $\mathbf{x}_0$ of a curve $\mathbf{x}=\mathbf{x}(s)$, where s is the distance along the curve, and then transport these vectors so that they do not rotate about $\hat{b}$. The required transport equation is

$$\frac{d\hat{\tau}_i}{ds} = -\hat{b}\left(\frac{\partial\hat{b}}{\partial s}\cdot\hat{\tau}_i\right), \tag{91}$$

for $i=1, 2$; the orthonormal nature of $(\hat{\tau}_1, \hat{\tau}_2, \hat{b})$ is preserved under this transport, and we also have $d\alpha/ds = \hat{\tau}_2\cdot d\hat{\tau}_1/ds = 0$. This transport process defines $\hat{\tau}_1$ and $\hat{\tau}_2$ along the given curve, and by using different curves emanating from $\mathbf{x}_0$ we can define $\hat{\tau}_1$ and $\hat{\tau}_2$ at all points of space.

A remarkable fact emerges, however, when we consider the transport of $\hat{\tau}_1$ and $\hat{\tau}_2$ around a closed curve: the perpendicular unit vectors do not return to their original values, but rather suffer a rotation. Let us consider an infinitesimal loop represented by the area vector $d\mathbf{A}$, and take the transport around the loop at fixed time, for simplicity. Then, we find that the net rotation angle $d\alpha$ about $\hat{b}$ is $-\mathbf{N}\cdot d\mathbf{A}$, where the vector N is implicitly given by

$$\epsilon_{ijk}N_k = \epsilon_{lnm}b_l b_{i,n}b_{j,m}\,, \tag{92}$$

in which $b_{i,j}$ means $\partial b_i/\partial x_j$. In general, $\mathbf{N}\neq 0$. Thus, we see that if Eq. (91) is used to define $\hat{\tau}_1$ and $\hat{\tau}_2$ as vector fields, then the values of these vectors at a point is dependent on the path used to reach that point. In general, it is impossible to define $\hat{\tau}_1$ and $\hat{\tau}_2$ so that they do not rotate about $\hat{b}$ along all possible paths. In this sense the last two terms of Eq. (89) are inevitable; they certainly are not an idiosyncrasy of the Hamiltonian methods of this paper.

We remark that the phenomenon described here has an almost perfect formal analogy to Thomas precession,[23] which is well-known in atomic physics. The analog of our transport equation (91) is Fermi–Walker transport.[24]

If we accept these facts and simply define $\hat{\tau}_1$ and $\hat{\tau}_2$ as fields in an arbitrary way, then another interesting conclusion emerges from a simple analysis, namely $\mathbf{N} = \nabla\times(\nabla\hat{\tau}_1\cdot\hat{\tau}_2) = \nabla\times\mathbf{R}$. This is proved in Appendix B. Note that although R depends on the definition of the perpendicular unit vectors, N depends only on $\hat{b}$. Stokes' theorem applied to this equation gives another interpretation of the vectors R and N in terms of the accumulated gyrophase around a closed loop, due to purely geometrical effects.

The other term in Eq. (91), $\hat{b}\cdot\nabla\times(\overline{U}\hat{b}+\mathbf{D})$, has a more conventional interpretation, which we will briefly describe. As the particle gyrates, the value of $\hat{b}$ as seen by the particle differs by terms which are first order in the gyroradius from the value of $\hat{b}$ at the guiding center. If we use the latter to define an average perpendicular plane, then $\hat{b}$ as seen by the particle will undergo a nutation, or tilting back and forth, giving it a periodic, small component in the perpendicular plane. This causes the instantaneous parallel velocity $u\hat{b}$ to enter into the reckoning of the perpendicular velocity relative to the average perpendicular plane, and hence, by Eq. (9d), into the gyrophase as well. The overall effect has a nonzero average, which a little geometry and a simple calculation shows to be $\frac{1}{2}\overline{U}\hat{b}\cdot(\nabla\times\hat{b})$. The term $\frac{1}{2}\hat{b}\cdot(\nabla\times\mathbf{D})$ has a similar explanation.

We can now give a geometrical interpretation to Eq. (84). In this equation, ψ is determined so that the rate of rotation of the gyrophase along the magnetic field lines due to the gradient in ψ exactly cancels the rate due to the nutation of $\hat{b}$ and the twisting of $\hat{\tau}_1$ and $\hat{\tau}_2$. In the case of purely magnetostatic fields, the lowest order particle drift $\overline{U}\hat{b}$ is also along field lines, i.e., $\mathbf{D}=0$, and Eq. (84) causes the $O(1)$ terms of $d\bar{\theta}/dt$ to completely cancel.

V. A NEW SET OF AVERAGED COORDINATES

The averaged variables $(\overline{\mathbf{X}}, \overline{K}, \overline{U}, \bar{\theta})$ of Sec. IV depend on the original choice of the perpendicular unit vectors, as may be seen by picking out the terms involving R and S in Eqs. (82). So also do the Hamiltonian (77) and the equations of motion, although the Poisson brackets, of Eqs. (33) do not. It can be shown that if we demand that the Poisson tensor be block diagonal, as in Eqs. (33), then the averaged variables $(\overline{\mathbf{X}}, \overline{K}, \overline{U}, \bar{\theta})$ cannot be freed of their dependence on R and S by a redefinition of variables.

However, if we sacrifice the demand that the Poisson tensor be block-diagonal, then we can find a coordinate

30 October 2026 19:36:15

system in which only the gyrophase $\bar{\theta}$ depends on the perpendicular unit vectors. This is the irreducible minimum, because, according to the discussion of Sec. IVE, the variable $\bar{\theta}$ cannot by any means be freed of its dependence on the perpendicular unit vectors, nor can its equation of motion. In the new coordinates the Poisson brackets involving $\bar{\theta}$ will depend on the perpendicular unit vectors, but the other variables, their equations of motion, the Poisson brackets among them, and the Hamiltonian will not.

Referring to Eqs. (16)–(17), we see that the F_n and G_n which depend on **R** and S are F_0, F_1, F_2, F_9, and G_7. Looking for the occurrence of these in Eqs. (82), we find

$$\begin{aligned}
\bar{\mathbf{X}} &= \cdots + (\epsilon^2 w^2/2B^2)(\hat{a}\hat{c} - \hat{c}\hat{a})\cdot\mathbf{R} + O(\epsilon^3)\,,\\
\bar{K} &= \cdots + (\epsilon w^2/2B)(\hat{a}\cdot\nabla\mathbf{D}\cdot\hat{c} + \mathbf{D}\cdot\mathbf{R} + S) + O(\epsilon^2)\,,\\
\bar{U} &= \cdots - (\epsilon w^2/2B)(\hat{a}\cdot\nabla\hat{b}\cdot\hat{c} + \hat{b}\cdot\mathbf{R}) + O(\epsilon^2)\,,\\
\bar{\theta} &= \theta + \cdots - (\epsilon w/B)\hat{a}\cdot\mathbf{R} + O(\epsilon^2)\,,
\end{aligned} \tag{93}$$

where the ellipsis indicates terms which are independent of **R** or S.

If we wish to have averaged guiding center variables which are independent of **R** or S, we can try to cancel the offensive terms of Eqs. (93) by adding functions of the averaged variables $(\bar{\mathbf{X}}, t, \bar{K}, \bar{U}, \bar{M})$. A function of $\bar{\theta}$ is not allowed, since it would bring back rapid oscillations. This procedure works for $(\bar{\mathbf{X}}, \bar{K}, \bar{U})$, and we are led to define the following primed variables:

$$\begin{aligned}
\bar{\mathbf{X}}' &= \bar{\mathbf{X}} - (\epsilon^2\bar{M}/B)\hat{b}\times\mathbf{R}\,,\\
\bar{K}' &= \bar{K} - \epsilon\bar{M}[\tfrac{1}{2}\hat{b}\cdot(\nabla\times\mathbf{D}) + \mathbf{D}\cdot\mathbf{R} + S]\,,\\
\bar{U}' &= \bar{U} + \epsilon\bar{M}[\tfrac{1}{2}\hat{b}\cdot(\nabla\times\hat{b}) + \hat{b}\cdot R]\,.
\end{aligned} \tag{94}$$

The procedure does not work for $\bar{\theta}$, and we leave it unchanged. In this manner we arrive at a new coordinate system of averaged variables on phase space, denoted by $(\bar{\mathbf{X}}', t, \bar{K}', \bar{U}', \bar{\theta}, \bar{M})$.

It is possible to compute the Poisson tensor in the new coordinates directly from Eqs. (94), using the chain rule formula for Poisson brackets. But, the calculation is quite tedious, and it is considerably easier to proceed via the action form ρ and the Lagrange tensor ω.

Equation (59) gives the components of ρ in a form which is valid for the variables $(\bar{\mathbf{X}}, t, \bar{K}, \bar{U}, \bar{\theta}, \bar{M})$. It is but a short calculation to transform these to the new, primed coordinates, using the transformation law for covariant vectors. The result is

$$\begin{aligned}
&\rho_{\bar{\mathbf{X}}'} = \frac{1}{\epsilon}\mathbf{A}^{**} + \epsilon\nabla W\,,\quad \rho_t = -K' - \frac{1}{\epsilon}\phi^{**} + \epsilon\frac{\partial W}{\partial t}\,,\\
&\rho_{\bar{U}'} = \rho_{\bar{K}'} = 0\,,\quad \rho_{\bar{\theta}} = \epsilon\bar{M}\,,\quad \rho_{\bar{M}} = \epsilon\frac{\partial W}{\partial\bar{M}}\,,
\end{aligned} \tag{95}$$

where

$$W = (\bar{M}/B)\mathbf{A}\cdot(\hat{b}\times\mathbf{R})\,,$$

and where the double-starred variables form a new set of modified potentials

$$\begin{aligned}
\mathbf{A}^{**} &= \mathbf{A} + \epsilon(\bar{U}'\hat{b} + \mathbf{D}) - \epsilon^2\bar{M}[\mathbf{R} + \tfrac{1}{2}\hat{b}\hat{b}\cdot(\nabla\times\hat{b})]\,,\\
\phi^{**} &= \phi + \frac{\epsilon}{2}D^2 + \epsilon^2\bar{M}[S + \tfrac{1}{2}\hat{b}\cdot(\nabla\times\mathbf{D})]\,.
\end{aligned} \tag{96}$$

All fields on the right-hand sides of Eqs. (95) and (96) are evaluated at $(\bar{\mathbf{X}}', t)$, and ∇ means $\partial/\partial\bar{\mathbf{X}}'$. The scalar W contributes to Eqs. (95) only through its differential, i.e., phase space gradient, and the terms containing it may be discarded without affecting the Lagrange or Poisson tensors [see Eq. (21)].

The double-starred potentials of Eq. (96) are evidently the next in sequence, after Eqs. (27)–(28), of a systematic development. From $\mathbf{A}^{**}$ and ϕ^{**} we define modified fields exactly as we did in Eqs. (29). Although **R** and S appear in the new modified potentials, the modified fields $\mathbf{E}^{**}$ and $\mathbf{B}^{**}$ involve only such derivatives of **R** and S as are independent of the perpendicular unit vectors [see Eqs (B15)–(B16)].

The Lagrange tensor is easily derived from Eqs. (95). Most of the new components are obtained by simply replacing the starred variables of Eqs. (32) with double-starred variables, and by evaluating at the new, primed coordinates. In addition, we have the following components of the Lagrange tensor, the counterparts of which in the variables $(\mathbf{X}, t, K, U, \theta, M)$ were zero

$$\begin{aligned}
\omega_{\bar{\mathbf{X}}'\bar{M}} &= \epsilon[\mathbf{R} + \tfrac{1}{2}\hat{b}\hat{b}\cdot(\nabla\times\hat{b})]\,,\\
\omega_{t\bar{M}} &= \epsilon[S + \tfrac{1}{2}\hat{b}\cdot(\nabla\times\mathbf{D})]\,.
\end{aligned} \tag{97}$$

Thus, the Lagrange tensor is no longer block-diagonal.

The inversion of ω_{ij} to find the Poisson tensor is trivial using the reduction rule of Appendix C and the previously derived Poisson brackets of Eqs. (33). Equations (33) go over to the new, primed variables simply by adding an additional star to all the modified fields and evaluating at the primed coordinates. In addition, the Poisson brackets containing $\bar{\theta}$ are nonzero:

$$\{\bar{\mathbf{X}}', \bar{\theta}\} = (\epsilon/B)\hat{b}\times\mathbf{R} + O(\epsilon^2)\,, \tag{98a}$$

$$\{\bar{K}', \bar{\theta}\} = S + \mathbf{D}\cdot\mathbf{R} + \tfrac{1}{2}\hat{b}\cdot(\nabla\times\mathbf{D}) + O(\epsilon)\,, \tag{98b}$$

$$\{\bar{U}', \bar{\theta}\} = -\hat{b}\cdot\mathbf{R} - \tfrac{1}{2}\hat{b}\cdot(\nabla\times\hat{b}) + O(\epsilon)\,. \tag{98c}$$

The Hamiltonian in the new, primed coordinates is easily found by substituting Eqs. (94) into Eq. (77). It is

$$\bar{H}(\bar{\mathbf{X}}', t, \bar{K}', \bar{U}', \bar{M}) = \bar{M}B + \tfrac{1}{2}\bar{U}'^2 - \bar{K}' + O(\epsilon^2)\,. \tag{99}$$

The $O(\epsilon)$ term vanishes completely, and this allows the equations of motion to be written very compactly.

The time evolution of $\bar{\theta}$ follows from Eqs. (98) and (99), and it is the same as given previously in Eq. (89). The equations of motion for the other variables are given in Appendix D.

VI. CONCLUSIONS

We have seen that a strict adherence to a block-diagonal Poisson tensor leads to guiding center variables which depend on the perpendicular unit vectors, but that all this dependence can be confined to the gyrophase if we relax the requirement for a block-diagonal Poisson tensor. The resulting set of variables, the

30 October 2026 19:36:15

primed variables of Sec. V, are in many ways the nicest set of all. Furthermore, the program of Ref. 1 of iterating the Darboux algorithm is not hindered by the off-diagonal elements; one can simply apply the Darboux algorithm to a reduced phase space of six dimensions, with coordinates $(\bar{\mathbf{X}}', t, \bar{K}', \bar{U}')$, in which $\bar{M}$ appears only as a parameter. The primed coordinates of Sec. V could never have been discovered in an approach which uses canonical coordinates, because these have the most rigidly block-diagonal Poisson tensor possible.

Nevertheless, it would be more satisfactory if the primed coordinates of Sec. V could be reached directly, instead of passing through the intermediaries of Secs. III and IV. The requirements of Sec. III, which demand vanishing Poisson brackets with θ, are at odds with what we now suspect is the desirable result, namely, the off-diagonal Poisson brackets involving $\bar{\theta}$. Further research on this question is called for; we believe that an approach which focuses on the action form ρ, and directly produces the sequence of modified fields indicated by Eqs. (27) and (96), would be the most effective.

Several applications of these methods are possible in the area of single particle motion. First, we can now regard the guiding center system as a solved problem, and consider additional perturbations. A high-frequency electromagnetic wave is an important example; this case has been studied by Grebogi, Kaufman, and Littlejohn.[25]

Second, we can iterate the Darboux algorithm and introduce the bounce phase. If this is done after we have averaged over the gyrophase, then we obtain the adiabatic invariant series for the bounce motion and averaged equations for motion across field lines. We have actually carried out this calculation, and our results agree with the older work of Northrop, Kruskal, and Liu[21] and Hastie, Taylor, and Haas.[22] If the bounce phase is introduced before we average over gyrophase, then we can average over both phases at once, and study the resonance structure. Although much work has been done on this problem in the past, a complete and systematic treatment has yet to be given.

Third, there is considerable interest at present in the effects of symmetries, especially axial symmetry, and of small symmetry breaking.[12] It is quite straightforward to build a symmetry into the formalism of this paper. In fact, the calculations are considerably simplified, because the number of degrees of freedom is reduced.

In the area of collective effects, the possible applications are too numerous to mention. We simply point out that the use of Hamiltonian methods is an effective way of organizing complicated calculations and of seeing important structure and symmetry in the results.

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APPENDIX A: HOW TO SPECIALIZE TO $\mathbf{E} = \partial/\partial t = 0$

Guiding center motion in purely magnetostatic fields is an important special case in its own right. The few aspects about specializing to this case that are not obvious are discussed here.

The main difference between the special magnetostatic case and the general case is that in the former we need only a six-dimensional phase space, with coordinates $(\mathbf{q}, \mathbf{p})$, $(\mathbf{x}, \mathbf{v})$, $(\mathbf{x}, u, \theta, w)$, $(\mathbf{X}, U, \theta, M)$, $(\bar{\mathbf{X}}, \bar{U}, \bar{\theta}, \bar{M})$, or $(\bar{\mathbf{X}}', \bar{U}', \bar{\theta}, \bar{M})$. The coordinates h, k_0, k, K, $\bar{K}$, and $\bar{K}'$ are not needed, and any equation defining them may be deleted in the special case. Likewise, the time t is no longer a phase coordinate, but rather is used in the usual way as the parameter of the phase trajectories. In the special case, any equation giving the components of the Poisson tensor, the Lagrange tensor, or the action form for the suppressed coordinates is not needed and may be deleted. The various Hamiltonians, (1), (5), (13), (44), (45), (67), (77), and (99), no longer depend on the suppressed coordinates t or h, k_0, k, K, $\bar{K}$, or $\bar{K}'$, and the terms on the right-hand sides of these equations, $-h$, $-k_0$, etc., are simply deleted. In the magnetostatic case, the Hamiltonians represent the physical energy of the particle, which is strictly conserved by the equations of motion. There are no nonphysical constants of motion.

APPENDIX B: VECTOR IDENTITIES

Both $(\hat{\tau}_1, \hat{\tau}_2, \hat{b})$ and $(\hat{a}, \hat{b}, \hat{c})$ are orthonormal triads, the latter being dependent on θ as well as $(\mathbf{x}, t)$ [see Eq. (11)]. The following identities are elementary:

$$\hat{\tau}_1\hat{\tau}_1 + \hat{\tau}_2\hat{\tau}_2 + \hat{b}\hat{b} = \hat{c}\hat{c} + \hat{a}\hat{a} + \hat{b}\hat{b} = \mathsf{I}\,, \tag{B1}$$

$$(\hat{\tau}_2\hat{\tau}_1 - \hat{\tau}_1\hat{\tau}_2)\cdot\mathbf{V} = (\hat{a}\hat{c} - \hat{c}\hat{a})\cdot\mathbf{V} = \hat{b}\times\mathbf{V}\,, \tag{B2}$$

$$\hat{c}\cdot\nabla\mathbf{V}\cdot\hat{a} - \hat{a}\cdot\nabla\mathbf{V}\cdot\hat{c} = \hat{b}\cdot(\nabla\times\mathbf{V})\,, \tag{B3}$$

for any vector $\mathbf{V}$.

If $(\hat{e}_1, \hat{e}_2, \hat{e}_3)$ represents either triad, then orthonormality implies

$$\nabla\hat{e}_i\cdot\hat{e}_j = -\nabla\hat{e}_j\cdot\hat{e}_i\,, \tag{B4}$$

$$\nabla\hat{e}_i\cdot\hat{e}_i = 0 \quad \text{(no sum on } i)\,. \tag{B5}$$

We normally deal with the fields $\mathbf{D}$ and $E_\parallel$ instead of $\mathbf{E}$. The relation is

$$\mathbf{E} = B\hat{b}\times\mathbf{D} + E_\parallel\hat{b}\,. \tag{B6}$$

The homogeneous Maxwell equations are

$$\nabla\cdot\hat{b} = -\hat{b}\cdot\nabla B/B\,, \tag{B7}$$

$$\frac{\partial\hat{b}}{\partial t} = -\hat{b}\left(\frac{1}{B}\frac{\partial B}{\partial t} + \frac{\mathbf{D}\cdot\nabla B}{B} + \nabla\cdot\mathbf{D}\right) + \hat{b}\cdot\nabla\mathbf{D} - \mathbf{D}\cdot\nabla\hat{b} + \frac{1}{B}\left(-E_\parallel\nabla\times\hat{b} + \hat{b}\times\nabla E_\parallel\right). \tag{B8}$$

30 October 2026 19:36:15

The following identities result from expanding the double cross product in two different ways:

$$\hat{b}\times[\hat{b}\times(\nabla\times\hat{b})]=-\hat{b}\times(\hat{b}\cdot\nabla\hat{b})=\hat{b}(\hat{b}\cdot\nabla\times\hat{b})-\nabla\times\hat{b}\,, \tag{B9}$$

$$\hat{b}\times[\mathbf{D}\times(\nabla\times\hat{b})]=\hat{b}\times[\nabla\hat{b}\cdot\mathbf{D}-\mathbf{D}\cdot\nabla\hat{b}]=\mathbf{D}(\hat{b}\cdot\nabla\times\hat{b})\,, \tag{B10}$$

$$\hat{b}\times[\hat{b}\times(\nabla\times\mathbf{D})]=\hat{b}\times[\nabla\mathbf{D}\cdot\hat{b}-\hat{b}\cdot\nabla\mathbf{D}]=\hat{b}(\hat{b}\cdot\nabla\times\mathbf{D})-\nabla\times\mathbf{D}\,. \tag{B11}$$

The normalization of $\hat{b}$ implies

$$\hat{b}\times(\nabla\times\hat{b})=-\hat{b}\cdot\nabla\hat{b}\,, \tag{B12}$$

$$\mathbf{V}\cdot\nabla\nabla\hat{b}\cdot\hat{b}=-\nabla\hat{b}\cdot(\mathbf{V}\cdot\nabla\hat{b})\,, \tag{B13}$$

for any vector V.

We have the following differential equations in θ:

$$\frac{\partial\hat{a}}{\partial\theta}=\hat{c}\,,\quad \frac{\partial\hat{c}}{\partial\theta}=-\hat{a}\,,\quad \frac{\partial Q_0}{\partial\theta}=-2Q_1\,,$$
$$\frac{\partial Q_1}{\partial\theta}=2Q_0\,,\quad \frac{\partial P_0}{\partial\theta}=\frac{\partial P_1}{\partial\theta}=0\,. \tag{B14}$$

[See Eqs. (75) for a definition of the tensors P_0, P_1, Q_0, Q_1.]

The vector $\mathbf{R}=\nabla\hat{c}\cdot\hat{a}$ and the scalar $S=\hat{a}\cdot\partial\hat{c}/\partial t$ satisfy the following identifies:

$$\mathbf{N}\equiv\nabla\times\mathbf{R}=\tfrac{1}{2}\hat{b}[(b_{i,j}b_{j,i})-(\nabla\cdot\hat{b})^2]$$
$$+(\nabla\cdot\hat{b})\hat{b}\cdot\nabla\hat{b}-\hat{b}\cdot\nabla\hat{b}\cdot\nabla\hat{b}\,, \tag{B15}$$

$$-\frac{\partial\mathbf{R}}{\partial t}+\nabla S=\nabla\hat{b}\cdot\left(\hat{b}\times\frac{\partial\hat{b}}{\partial t}\right), \tag{B16}$$

[see also Eqs. (73) and (92).] Equation (B15) is proved by directly computing $\nabla\times(\nabla\hat{c}\cdot\hat{a})$, inserting the identity tensor (B1), and using Eqs. (B4)–(B5). Equation (B16) is proved similarly. These equations are used in deriving the results of Appendix D.

APPENDIX C: INVERTING AN ANTISYMMETRIC MATRIX

An $n\times n$, antisymmetric matrix A is singular if n is odd. If $n=2N$ is even, then $\det(A)$ is nonnegative.

The inversion of many of the $2N\times2N$, antisymmetric matrices of this paper can be reduced to the inversion of a $(2N-2)\times(2N-2)$ matrix by means of the following reduction rule. If the rows and columns of A can be arranged (typically by permuting the corresponding coordinates) so that A has the form

$$A=\begin{pmatrix} C & 0 & X \\ 0 & 0 & -r \\ -X & r & 0 \end{pmatrix}, \tag{C1}$$

where C is a $(2N-2)\times(2N-2)$ matrix, X is a $(2N-2)$-vector, and r is a scalar, then $B=A^{-1}$ can be written

$$B=\begin{pmatrix} D & Y & 0 \\ -Y & 0 & s \\ 0 & -s & 0 \end{pmatrix}, \tag{C2}$$

where $D=C^{-1}$, $Y=(1/r)C^{-1}X$, and $s=1/r$. Some of the matrices of this paper can be reduced twice by this rule.

We write A_{ij} or $A(i,j)$ impartially for the components of a $2N\times2N$, antisymmetric matrix A. We define the following quantity:

$$\Delta=\sum_P S_P A(i_1,j_1)A(i_2,j_2)\cdots A(i_N,j_N)\,, \tag{C3}$$

where the sum is taken over all permutations $(1,2,\ldots,2N)\to(i_1,j_1,i_2,j_2,\ldots,i_N,j_N)$ which satisfy $i_1<i_2<\cdots<i_N$, and $i_k<j_k$ for all k, and where S_P is the parity of the permutation. For example, if A is a 4×4 matrix,

$$\Delta=A_{12}A_{34}-A_{13}A_{24}+A_{14}A_{23}\,. \tag{C4}$$

The sum in Eq. (C3) has $(2N-1)!!$ terms.

The determinant of A is Δ^2. If $\Delta\neq0$, then $B=A^{-1}$ is given by the following rule. If $i>j$, set $B_{ij}=-B_{ji}$. If $i<j$, then

$$B_{ij}=\frac{(-1)^{i+j}}{\Delta}\sum_P S_P A(i_1,j_1)$$
$$\times A(i_2,j_2)\cdots A(i_{N-1},j_{N-1})\,, \tag{C5}$$

where the sum is taken over all permutations of the numbers $(1,2,\ldots,2N)$, with i,j removed, to $(i_1,j_1,\ldots,i_{N-1},j_{N-1})$, where $i_1<i_2<\cdots<i_{N-1}$ and $i_k<j_k$ for $k=1,2,\ldots,N-1$. Again, S_P is the parity of the permutation. The sum (C5) has $(2N-3)!!$ terms.

These formula represent the simplification to Cramer's rule which comes about because of antisymmetry. They may easily be proved by induction.

APPENDIX D: HIGHER ORDER DRIFT EQUATIONS

In this appendix we give the higher order drift equations in a form which would be suitable for numerical integration, and which also displays some of the structure discovered in the main text.

The guiding center variables of this appendix are $(\mathbf{X},U,M)$. These are identical to the variables $(\bar{\mathbf{X}}',\bar{U}',\bar{M})$ of the main text [see Eqs. (94)]. The overbar and primes have been dropped for convenience, and the variables of this appendix are not to be confused with the intermediate guiding center variables of Sec. III. Other notational differences are clear from the context or will be noted as they arise.

We choose units so that $e=m=c=1$. See Sec. II concerning restoring physical units. We assume $\mathbf{B}=\mathbf{E}_\perp=O(1)$, and $E_\parallel=O(\epsilon)$.

Let x and v be the position and velocity of the particle, let $\hat{b}$ be the unit vector along the magnetic field B, and let $\mathbf{D}=\mathbf{E}\times\mathbf{B}/B^2$ be the $\mathbf{E}\times\mathbf{B}$ drift velocity. Let $u=\hat{b}\cdot\mathbf{v}$ be the parallel velocity of the particle, and let $\mathbf{v}_a=\hat{b}\times(\mathbf{v}-\mathbf{D})$, $\mathbf{v}_c=-\hat{b}\times\mathbf{v}_a$, and let $w=|\mathbf{v}_c|=|\mathbf{v}_a|$. We define the scalars b_0 and D_0 by

$$b_0=\hat{b}\cdot(\nabla\times\hat{b})\,,\quad D_0=\hat{b}\cdot(\nabla\times\mathbf{D})\,. \tag{D1}$$

Then the guiding center position X is given by

$$\mathbf{X}=\mathbf{x}-(\epsilon/B)\mathbf{v}_a+(\epsilon^2/B^2)(R_{2a}\mathbf{v}_a+R_{2b}\hat{b})+O(\epsilon^3)\,, \tag{D2}$$

30 October 2026 19:36:15

where

$$R_{2a} = D_0 + ub_0 - (1/2B)\mathbf{v}_a \cdot \nabla B,$$

$$R_{2b} = \mathbf{v}_c \cdot \left(\hat{b} \cdot \nabla \mathbf{D} + \mathbf{D} \cdot \nabla \hat{b} + \frac{\partial \hat{b}}{\partial t} + 2u\hat{b} \cdot \nabla \hat{b}\right) + \tfrac{1}{8}(\mathbf{v}_c \cdot \nabla \hat{b} \cdot \mathbf{v}_c - 5\mathbf{v}_a \cdot \nabla \hat{b} \cdot \mathbf{v}_a). \quad \text{(D3)}$$

We allow the operator ∇ to act only on the immediately following operand. For example, $\nabla \hat{b}$ plays the role of a matrix A with components $A_{ij} = \partial b_j / \partial x_i$. The quantity ϵ is a dimensionless ordering parameter; it should be set to unity in applications.

The parallel velocity of the guiding center U is given by

$$U = u - (\epsilon/B)\{\mathbf{v}_a \cdot [u\hat{b} \cdot \nabla \hat{b} - \hat{b} \times (\nabla \times \mathbf{D})] + \tfrac{1}{4}(3\mathbf{v}_a \cdot \nabla \hat{b} \cdot \mathbf{v}_c - \mathbf{v}_c \cdot \nabla \hat{b} \cdot \mathbf{v}_a)\} + O(\epsilon^2). \quad \text{(D4)}$$

Finally, the magnetic moment M is given by

$$M = \frac{w^2}{2B} + \frac{\epsilon}{B^2}\left\{\left[\left(u\hat{b} \cdot \nabla + \mathbf{D} \cdot \nabla + \frac{\partial}{\partial t}\right)(u\hat{b} + \mathbf{D})\right] \cdot \mathbf{v}_a + \tfrac{1}{4}(3\mathbf{v}_a \cdot \nabla \mathbf{D} \cdot \mathbf{v}_c - \mathbf{v}_c \cdot \nabla \mathbf{D} \cdot \mathbf{v}_a) + (u/4)(3\mathbf{v}_a \cdot \nabla \hat{b} \cdot \mathbf{v}_c - \mathbf{v}_c \cdot \nabla \hat{b} \cdot \mathbf{v}_a) + \frac{w^2}{2B}\mathbf{v}_a \cdot \nabla B\right\} + O(\epsilon^2). \quad \text{(D5)}$$

All quantities on the right-hand sides of Eqs. (D2)–(D5) are evaluated at the instantaneous particle position. There are different ways to write these definitions, using various vector identities. The forms given may be regarded as more or less arbitrary. On the basis of averaging alone, the higher order terms of (D2) and (D4) are not unique. However, we believe that these definitions are especially appropriate, because they allow simple forms for the equations of motion.

The guiding center equations of motion are most compactly expressed in terms of the modified fields $\mathbf{B}^*$ and $\mathbf{E}^*$. (These are the double-starred fields of the main text.) These are evaluated at the guiding center position, and they are given by

$$\mathbf{B}^* = \mathbf{B} + \epsilon(U\nabla \times \hat{b} + \nabla \times \mathbf{D}) + (\epsilon^2 M/2)\{\hat{b}[(\nabla \cdot \hat{b})^2 - (b_{i,j}b_{j,i})] - 2(\nabla \cdot \hat{b})\hat{b} \cdot \nabla \hat{b} + 2\hat{b} \cdot \nabla \hat{b} \cdot \nabla \hat{b} - \nabla \times (b_0 \hat{b})\}, \quad \text{(D6)}$$

$$\mathbf{E}^* = \mathbf{E} - \epsilon\left(U\frac{\partial \hat{b}}{\partial t} + \frac{\partial \mathbf{D}}{\partial t} + \nabla \mathbf{D} \cdot \mathbf{D}\right) + \frac{\epsilon^2 M}{2}\left[-2\nabla \hat{b} \cdot \left(\hat{b} \times \frac{\partial \hat{b}}{\partial t}\right) + \frac{\partial}{\partial t}(b_0 \hat{b}) - \nabla D_0\right]. \quad \text{(D7)}$$

In Eq. (D6), $b_{i,j}$ means $\partial b_i / \partial x_j$, and the summation convention is employed. All of the complexity of the equations of motion is contained in these variables. The fields given satisfy the pair of homogeneous Maxwell equations, with $\mathbf{E}$ and $\mathbf{B}$ replaced by $\mathbf{E}^*$ and $\mathbf{B}^*$.

In addition to $\mathbf{E}^*$ and $\mathbf{B}^*$, we define the scalar B^*:

$$B^* = \hat{b} \cdot \mathbf{B}^* = B + \epsilon(Ub_0 + D_0) + (\epsilon^2 M/2)[\nabla^2 \hat{b} \cdot \hat{b} + (\nabla \cdot \hat{b})^2 + (\hat{b} \cdot \nabla \hat{b})^2]. \quad \text{(D8)}$$

Note that $B^* \neq |\mathbf{B}^*|$.

The equations of motion for the variables $\mathbf{X}$ and U have the following form:

$$\frac{d\mathbf{X}}{dt} = \frac{1}{B^*}(\mathbf{E}^* \times \hat{b} + U\mathbf{B}^* + \epsilon M\hat{b} \times \nabla B), \quad \text{(D9)}$$

$$\frac{dU}{dt} = \frac{\mathbf{B}^*}{B^*} \cdot \left(\frac{\mathbf{E}^*}{\epsilon} - M\nabla \mathbf{B}\right). \quad \text{(D10)}$$

Equation (D10) and the parallel component of (D9) are valid through $O(\epsilon)$, and the perpendicular component of Eq. (D9) is valid through $O(\epsilon^2)$.

The familiar forms of the guiding center equations are power series in ϵ, whereas Eqs. (D9)–(D10) are rational functions of ϵ. It is straightforward to expand these out into power series, using Eq. (D8) and various vector identities (see Appendix B). By doing this, it is fairly easy to reproduce the usual drift equations:

$$\frac{d\mathbf{X}}{dt} = U\hat{b} + \mathbf{D} + \frac{\epsilon}{B}\hat{b} \times \left[M\nabla B + \left(U\hat{b} \cdot \nabla + \mathbf{D} \cdot \nabla + \frac{\partial}{\partial t}\right)(U\hat{b} + \mathbf{D})\right], \quad \text{(D11)}$$

$$\frac{dU}{dt} = \frac{E_\parallel}{\epsilon} - M\hat{b} \cdot \nabla B + \left[\left(U\hat{b} \cdot \nabla + \mathbf{D} \cdot \nabla + \frac{\partial}{\partial t}\right)\hat{b}\right] \cdot \mathbf{D}. \quad \text{(D12)}$$

These equations can be extended to one higher order. One result is easy, namely,

$$\left(\frac{d\mathbf{X}}{dt}\right)_\parallel = \hat{b}U + O(\epsilon^2). \quad \text{(D13)}$$

The $O(\epsilon)$ term vanishes, and U is indeed the parallel velocity of the guiding center. Likewise, Eq. (D12) and the perpendicular component of Eq. (D11) can be extended to one higher order, with the expenditure of more effort. However, the results are quite unwieldy and not very illuminating.

Instead, we feel that it is better to use Eqs. (D9) and (D10) in the form given. Not only are these equations practical for numerical work, but they also satisfy exactly two important identities which their expanded counterparts only satisfy approximately.

The first identity is conservation of energy. This applies, of course, only for time-independent fields. In that case, we define ϕ^* by

$$\phi^* = \phi + (\epsilon/2)D^2 + (\epsilon^2 M/2)D_0, \quad \text{(D14)}$$

where $\mathbf{E} = -\nabla\phi$. It is easily verified that $\mathbf{E}^* = -\nabla\phi^*$. Then, Eqs. (D9)–(D10) possess the following energy integral:

$$\frac{d}{dt}\left(MB + \frac{1}{2}U^2 + \frac{1}{\epsilon}\phi^*\right) = 0. \quad \text{(D15)}$$

The conserved quantity is the energy of the particle through $O(\epsilon)$.

The second identity is Liouville's theorem, which applies even for time-dependent fields. It has the form

$$\frac{\partial B^*}{\partial t} + \frac{\partial}{\partial \mathbf{X}} \cdot (B^*\dot{\mathbf{X}}) + \frac{\partial}{\partial U}(B^*\dot{U}) = 0. \quad \text{(D16)}$$

Further discussion of this equation may be found in Sec. III of the main text. It is straightforward to verify Eqs. (D15)–(D16) directly, using Eqs. (D9)–(D10).

30 October 2026 19:36:15

APPENDIX E: RELATIVISTIC MAGNETIC MOMENT SERIES

In this appendix we will give the first two terms of the magnetic moment series for relativistic guiding center motion.

We choose units so that $e=m=c=1$. We set $x^\mu=(t,\mathbf{x})$, for $\mu=0,1,2,3$, and we let the metric $g_{\mu\nu}$ have diagonal components $(-1,+1,+1,+1)$. We let $v^\mu=(\gamma,\gamma\boldsymbol{\beta})$ be the world velocity of the particle; γ and $\boldsymbol{\beta}$ are the usual relativistic variables. The magnetic moment is a function of x^μ and v^μ which is a formal constant of the motion. We will describe the construction of this function in a systematic manner.

The Lorentz scalars are

$$L_1=B^2-E^2\,,\quad L_2=\mathbf{E}\cdot\mathbf{B}\,. \tag{E1}$$

We assume that $L_1>0$ and $L_2=O(\epsilon)$. In terms of these we define two frequencies:

$$\Omega=[\tfrac{1}{2}(L_1^2+4L_2^2)^{1/2}+L_1]^{1/2}\,,$$
$$\nu=\mathrm{sgn}(L_2)[\tfrac{1}{2}(L_1^2+4L_2^2)^{1/2}-L_1]^{1/2}\,. \tag{E2}$$

Ω is the gyrofrequency with respect to proper time, and $1/\nu$ is the characteristic time to accelerate from rest to relativistic velocities along a field line. In the nonrelativistic limit, $\Omega=B$ and $\nu=E_\parallel$.

In terms of the electromagnetic field, we define the following two vectors:

$$e^\mu=(1/N)(E^2+\Omega^2,\mathbf{E}\times\mathbf{B})\,,$$
$$b^\mu=(1/N)(0,\nu\mathbf{E}+\Omega\mathbf{B})\,, \tag{E3}$$

where the normalization N is given by

$$N=[(\nu^2+\Omega^2)(E^2+\Omega^2)]^{1/2}. \tag{E4}$$

The vectors e^μ and b^μ are unit eigenvectors of the stress-energy tensor, and they satisfy $e^\mu e_\mu=-1$, $b^\mu b_\mu=+1$, $e^\mu b_\mu=0$. In the nonrelativistic limit e^μ is the time axis for a frame moving with the $\mathbf{E}\times\mathbf{B}$ drift velocity, and $b^\mu=(0,\hat{b})$, i.e., the spatial unit vector along the magnetic field.

In terms of the world velocity v^μ, we define the three scalars k, u, and w, and the two vectors v_a^μ and v_c^μ, as follows:

$$k=-e^\mu v_\mu\,,\quad u=b^\mu v_\mu\,,$$
$$v_c^\mu=v^\mu-ke^\mu-ub^\mu\,,$$
$$v_{a\mu}=-\epsilon_{\mu\nu\alpha\beta}e^\alpha b^\beta v_c^\nu\,, \tag{E5}$$
$$w=(v_c^\mu v_{c\mu})^{1/2}\,,$$

where the Levi-Civita tensor is specified by $\epsilon_{0123}=+1$. In the nonrelativistic limit, k is the kinetic energy plus rest mass of the particle, u is the parallel velocity, $v_c^\mu=(0,\mathbf{v}_\perp)$, and $v_a^C=(0,\hat{b}\times\mathbf{v}_\perp)$; both k and $\mathbf{v}_\perp$ are measured in a frame moving with the $\mathbf{E}\times\mathbf{B}$ drift.

In terms of these variables, the magnetic moment series is

$$M=\frac{w^2}{2\Omega}+\frac{\epsilon}{\Omega^2}\Big\{\frac{w^2}{2\Omega}(v_a^\mu\Omega_{,\mu})+\tfrac{1}{4}(3v_a^\mu v_{c\nu}-v_c^\mu v_{a\nu})(ke^\nu_{,\mu}+ub^\nu_{,\mu})$$
$$+[(ke^\mu+ub^\mu)(ke^\nu_{,\mu}+ub^\nu_{,\mu})]v_{a\nu}\Big\}+O(\epsilon^2)\,, \tag{E6}$$

where commas represent derivatives, e.g., $e^\nu_{,\mu}=\partial e^\nu/\partial x^\mu$, and where ϵ is a formal ordering parameter which is set to unity in applications. The nonrelativistic limit of this is shown in Eq. (D5).

The results of this appendix are indebted to the work of Fradkin.[26]

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30 October 2026 19:36:15